Characterization of Mixed Farming Systems in Malawi

Sustainable Intensification of Mixed Farming Systems (SI-MFS)

Initiative Baseline Evaluation Survey Report

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The Sustainable Intensification of Mixed Farming Systems Initiative aims to provide equitable, transformative pathways for improved livelihoods of actors in mixed farming systems through sustainable intensification within target agroecologies and socio-economic settings.

Through action research and development partnerships, the Initiative will improve smallholder farmers' resilience to weather-induced shocks, provide a more stable income and significant benefits in welfare, and enhance social justice and inclusion for 13 million people by 2030.

Activities will be implemented in six focus countries globally representing diverse mixed farming systems as follows: Ghana (cereal–root crop mixed), Ethiopia (highland mixed), Malawi: (maize mixed), Bangladesh (rice mixed), Nepal (highland mixed), and Lao People's Democratic Republic (upland intensive mixed/ highland extensive mixed).

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Executive Summary

The Sustainable Intensification of Mixed Farming Systems (SI-MFS) initiative is being implemented in six countries of the Global South: Ghana, Ethiopia, Malawi, Bangladesh, Nepal, and Laos. The initiative's main objective is to provide equitable, gender-transformative pathways for improving the livelihoods of smallholder farmers through action research. Interventions under the SI-MFS initiative focuses on five impact areas: (1) nutrition, health and food security; (2) poverty reduction, livelihoods and jobs; (3) gender equality, youth and social inclusion; (4) climate adaptation and mitigation; and (5) environmental health and biodiversity. By focusing on these impact areas, the initiative is contributing to several sustainable development goals (SDGs) including no poverty, zero hunger, good health, gender equality and empowering all women and girls, decent work, and economic growth.

This baseline report presents Malawi's SI-MFS data covering 1,269 farming households from six sampled districts: Mzimba in the Northern region, Dedza and Kasungu in the Central region, Balaka, Mangochi, and Zomba in the Southern part region. Of 1,269 farming households, 948 were treatment households and 321 were control households. Several characteristics including demography, agricultural land characteristics, gender, time use and labor, crop production, inputs and productivity, storage facility, livestock ownership, dwelling characteristics, agricultural-related shocks, food consumption expenditure, and non-food consumption expenditure were captured. Data analysis was done using several descriptive statistical tools including but not limited to cross-tabulations, contingency tables, and frequencies, presented in either graphical or tabular format.

With the current survey data, we found that crop production is the primary economic activity of the sampled households. Maize and legume crops such as groundnut, soybean, and pigeon peas are significant crops in the study districts. In addition to crop production, livestock farming is also a major farming activity, especially for animals such as chickens and goats. While most households have a diversified diet (they consume seven food groups), most (73%) are poor. Gender inequality on income and assets ownership including land, in particular among women and the youth is common. Even though some districts including Mzimba and Kasungu had more land under SI technologies such as conservation agriculture (i.e., crop rotation and zero/minimum tillage) and inorganic fertilizer, farmers in districts with higher population density such as Mangochi and Balaka allocated more land to intercropping, as a coping mechanism to land scarcity. Overall, food inflation (68%), rising agricultural input prices (67%), drought (53%) and crop pest and disease outbreaks (42%) are major agrarian shocks affecting farmers in the six study districts. Implications of study findings on the interventions under the SI-MFS initiative are provided.

List of Abbreviations

Africa RISING	Africa Research in Sustainable Intensification for Next Generation
ASF	Animal-Sourced Foods
CA	Conservation Agriculture
CGIAR	Consultative Group on International Agricultural Research
DAES	Department of Agricultural Extension Service
DeSIRA	Development of Smart Innovation through Research in Agriculture
EiA	Excellence in Agronomy
EPA	Extension Planning Areas
FCS	Food Consumption Score
GHG	Greenhouse Gas
HDDS	Household Dietary Diversity Score
ISP	Input Subsidy Program
MELIA	Monitoring Evaluation, Learning, and Impact Assessment
MFS	Mixed Farming Systems
MWK	Malawi Kwacha
PBF	Plant-Based Foods
PCF	Principal Component Factor
PICS	Purdue Improved Crop Storage Bags
PPS	Probability Proportional to Size Sampling
SGD	Sustainable Development Goals
SI	Sustainable Intensification
SI-MFS	Sustainable Intensification of Mixed Farming Systems
TLU	Tropical Livestock Units
UU	Ukama Ustawi
VSLA	Village Savings and Loan Associations

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1 | Introduction

With only seven growing seasons to achieve two critical sustainable development goals (SDGs): no poverty and zero hunger by 2030, the Global South still faces an enormous challenge to feed its current and future generations. While this challenge is associated with many factors, a few facets remain outstanding: (i) recurrent extreme weather events (Amadu et al., 2020; MacLaren et al., 2022; Tesfaye et al., 2021), (ii) global pandemics including Covid-19 (Amare et al., 2021; Headey & Ruel, 2022; Manyong et al., 2022; Osendarp et al., 2021), (iii) rising population growth (Ricker-Gilbert et al., 2014; United Nation, 2015; Willy et al., 2019), (iv) war, conflicts and migration (Deininger et al., 2023; Rao et al., 2019), and (v) low adoption of agricultural technologies, in particular, fertilizer and improved seeds (Khonje et al., 2022a; 2018; Manda et al., 2016). These factors probably explain why crop yields, including cereals and legumes, have remained low (<3.5 tons/ha) in Malawi (Benson, 2021; FAO, 2021; Khonje et al., 2022b).

To address these challenges, international research institutions and local partners are implementing different project activities on sustainable intensification (SI) of mixed farming systems (MFS) initiative in the six countries of the Global South: Ghana, Ethiopia, Malawi, Bangladesh, Nepal, and Laos. SI responds to the need to feed the growing populations by producing more food on the same piece of land and concurrently reducing environmental degradation (Garnett & Godfray, 2012). To identify context-specific pathways towards resilient, scalable MFS that preserve natural capital, an integrated systems research with SI is required.

More broadly, various components of MFS interact with each other and with the external environment, including climate and landscape (Auricht et al., 2019). One potential advantage of these farming systems is that they occur in almost all agro-ecological zones, with a diverse climatic and soil conditions and livelihood patterns (Dixon et al., 2019; Thornton et al., 2018). Moreover, not only is the MFS important because it covers 2.5 billion ha of land globally, but also because it supplys around 75% of milk, 60% of meat, and 41–86% of cereals consumed in the Global South (de Haan et al., 1997; Herrero et al., 2010).

The SI-MFS initiative aims to provide equitable, gender-transformative pathways for improving the livelihoods of smallholder farmers through several pathways. For instance, in an MFS, livestock provides animal-sourced foods (ASF), as these are rich in protein and micronutrients, and income to farming households, manure to fertilize the soil, and draft power for crop cultivation (Headey et al., 2018; Khonje et al., 2022a; Rawlins et al., 2014; Zaharia et al., 2021). However, increased consumption of ASF also has its downsides, as ASF have much larger climate and environmental footprints than plant-based foods (PBF) (Parlasca & Qaim, 2022; Springmann et al., 2016; Xu et al., 2021). While crops provide PBFs, income, organic manure, and livestock feed from crop residues (Khonje et al., 2022a; 2018; Manda et al., 2016; Tesfaye et al., 2021). Therefore, an integrated crop–livestock farming system has the potential to maintain ecosystem function and health, and to help prevent agricultural systems from becoming fragile by enhancing biodiversity and thus increasing capability to absorb shocks to the natural resource base (Holling, 1995; Peterson et al., 2020; Thornton et al., 2018).

Even though this (SI-MFS) initiative has several potential benefits to farmers, end consumers and environment, there are knowledge gaps on the biophysical and socio-economic interactions and dynamics (Dixon et al., 2019; Thornton et al., 2018). This can undermine many development-oriented interventions aiming at driving MFS towards SI. To address this knowledge gap, a baseline survey was conducted from December 2022 to January 2023 in Malawi. The objective of the study was to characterize SI-MFS. Therefore, this baseline report focuses on providing critical features of SI-MFS in Malawi.

Using cross-sectional data from the baseline survey, our descriptive analysis focuses on general farming characteristics and five impact areas of the initiative: (1) nutrition, health and food security; (2) poverty reduction, livelihoods and jobs; (3) gender equality, youth and social inclusion; (4) climate adaptation and mitigation; and (5) environmental health and biodiversity. Through different interventions under the SI-MFS initiative, outcomes from these impact areas will contribute to several SDGs including no poverty, zero hunger, good health, gender equality and empowering all women and girls, decent work, and economic growth. The Malawi baseline survey covered several topics: household demographics, time use and labor, agricultural land characteristics, gender, crop production, input use and productivity, storage facility, livestock ownership, dwelling characteristics, and asset ownership, food consumption expenditure, non-food consumption expenditure, and household shocks.

The baseline report is organized into four additional sections. Section 2 presents the contextual background, impact evaluation design, and sampling procedure. We then present detailed descriptive results on household demographics, agricultural land characteristics, crop production and input use, crop yields, storage facility, livestock ownership, dwelling characteristics, and asset ownership, and household shocks in section 3. Section 4, presents and discusses key results on the five impact areas of the SI-MFS initiative. The last section draws some conclusions.

2 | Methodology

2.1 | Selection of the survey sites

The SI-MFS initiative in Malawi is operating in six districts: Mzimba in the Northern region; Dedza and Kasungu in the Central region; Balaka, Mangochi, and Zomba in the Southern region (Figure 1). Several SI-MFS interventions focusing on different agricultural technologies are being implemented in these districts including conservation agriculture (CA), different types of intercropping (e.g., double-up legumes and *mbili mbili*) of other hybrid maize with improved groundnut, soybean, cowpea, pigeon pea, and bean varieties with different doses of inorganic fertilizer.

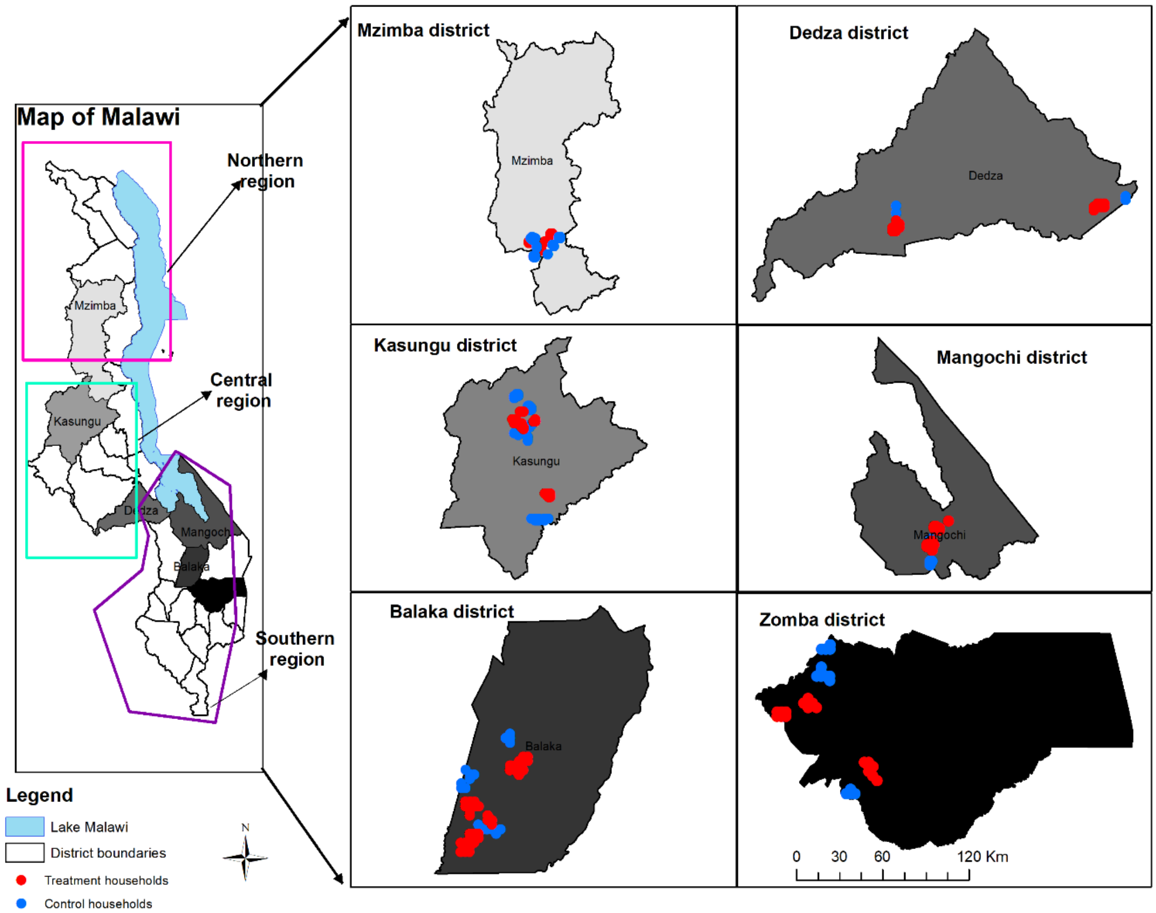


Figure 1 | Map of Malawi showing the sample distribution by survey district.

The initiative's six districts and extension planning areas (EPAs) were selected purposively following the criteria outlined below. The districts were first divided into three clusters: i.e., (i) Mzimba-Kasungu, (ii) Dedza, and (iii) Balaka-Mangochi-Zomba. The selection of the Mzimba-Kasungu cluster was guided by previous and ongoing CGIAR and other institutions' investments, such as the Development of Smart Innovation through Research in Agriculture (DeSIRA) project, Excellence in Agronomy (EiA) initiative on soybean use, and linkage with the Ukama Ustawi (UU) regional initiative. In addition, this cluster also has a good diversity of cropping systems and high livestock density, considering the importance of livestock in the initiative. The Dedza cluster was selected mainly because of the previous investments by the Africa Research in Sustainable Intensification for Next Generation (Africa RISING) project, and the existing functional partnerships. Similarly, the Balaka-Mangochi-Zomba cluster was chosen because of the current long-term collaboration with the Department of Agricultural Extension Service (DAES), the importance of livestock in the farming system, and co-location/synergies with the UU initiative sites.

Members of the SI-MFS initiative selected the treatment sections and villages within the selected EPAs. On the other hand, the Monitoring Evaluation, Learning, and Impact Assessment (MELIA) team set the control sections and villages. The sections were selected in such a way that the sections were similar as much as possible in terms of agroecology, population density, elevation, precipitation, temperature, market access, and farming systems, but far away to avoid contamination. Next, the villages in the control sections were selected using probability proportional to size (PPS) sampling.

2.2 | Power calculation and sampling procedure

A Quasi randomized control trial (QRCT) design was followed to obtain credible and robust impact estimates at the end of the initiative investment cycle. Since the treatment villages were purposively selected, implementing a cluster randomized design was impossible; instead, randomization was done at the household level. Using data from the Africa RISING end-line survey, we selected some technologies and their combinations, such as crop rotation, intercropping, inorganic fertilizer amendments, and improved crop varieties, and computed their adoption rates.

In general, the adoption rates of these technologies ranged from 2.3% to 42% on average (Shee et al., 2016). Using a baseline joint adoption of crop rotation, intercropping, and improved varieties of 6% (Shee et al., 2016), a 5-percentage point increase in adoption to 11%, and 80% power, a minimum sample size of 1,016 was deemed appropriate to detect the effect size of 0.05. To account for non-responses, the reserve sample was put at 25%. Ultimately, 1,269 households were sampled and interviewed (Table 1). In total, the survey covered 10 EPAs from the six districts. Of 1,269 households, 948 were treatment households, and 321 were control households.

Table 1 also shows the detailed geographical distribution of the sampled households (i.e., farming households) by district, EPA, and research group. Of 1,269 households, 320 were

Table 1: Distribution of sampled households by EPA and research group

District	EPA	Number of sections	Number of villages	Treatment households	Control households	Full sample
Balaka	Rivirivi	2	6	110	16	126
Balaka	Phalula	4	10	162	32	194
Dedza	Golomoti	2	4	55	15	70
Dedza	Lobi	2	6	107	17	124
Kasungu	Chulu	8	12	108	64	172
Kasungu	Lisasadzi	2	5	27	33	60
Mangochi	Nasenga	4	11	190	32	222
Mzimba	Champhira	6	11	108	64	172
Zomba	Masaula	4	6	54	32	86
Zomba	Dzaone	2	3	27	16	43
		36	74	948	321	1,269

from Balaka, where 126 were from Rivirivi EPA and 194 were from Phalura EPA. Specifically, the SI-MFS initiative covered two sections and six villages and four sections and ten villages in Rivirivi, and Phalura EPAs, respectively.

While 194 farming households were interviewed in Dedza, with 70 and 124 households coming from Golomoti and Lobi EPAs, respectively. In Kasungu, 97 control households and 135 treatment households were interviewed. Whereas 222, 172, and 129 households were interviewed for Mangochi, Mzimba and Zomba, respectively. While almost all the study districts covered more than one EPA, only one EPA (Champhira) was covered in Mzimba.

A sample size of 1,269 smallholder farming households was used in the analysis after data cleaning. Descriptive statistics (in tabular and graphical forms) were mainly used to summarize key features of sustainable intensification of mixed farming systems in Malawi.

3 | Results and Discussion

3.1 | Demographic characteristics

3.1.1 | Demographic characteristics by research group

Table 2 reports the summary statistics for selected demographic characteristics at the household level. Our analysis primarily focused on five characteristics: household size, adult years of education, adult age, age of the household head, and dependency ratio. The dependency ratio is a ratio between the number of people aged below 15 years and those above 64 years of age and the working population aged between the age of 15 and 64.

Table 2: Household level summary statistics by research group

	N	Mean	SD	Min	Max
Household size	1,269	5.02	2.07	1	17
Treatment	948	4.92***	2.05	1	17
Control	321	5.31	2.13	1	13
Average adult years of education	1,269	10.81	4.42	1	18
Treatment	948	10.58***	4.50	1	18
Control	321	11.49	4.09	1	18
Average adult years of age	1,269	37.85	13.07	18	85
Treatment	948	38.16	13.48	18	85
Control	321	36.91	11.77	18	80
HH head age in years	1,269	46.62	15.87	18	85
Treatment	948	46.79	16.08	18	85
Control	321	46.13	15.25	18	85
Dependency ratio	1,269	48.65	22.19	0	100
Treatment	948	48.80	22.73	0	100
Control	321	48.20	20.55	0	100
Percentage of HH head married or cohabiting	1,269	68.56	46.45	0	100
Treatment	948	67.30*	46.94	0	100
Control	321	72.27	44.83	0	100
Percentage of HH head female	1,269	33.02	47.05	0	100
Treatment	948	34.07	47.42	0	100
Control	321	29.91	45.86	0	100

Note: N, Number of observations (No. of farming households). SD, standard deviation. HH, Household head. ***, * Denotes the mean difference between treatment group and control group is significant at 1% and 10%, respectively.

The results show that the dependency ratio for the population is 48.65 on average (Table 2). The average household size is 5.02 members, with a range of 1 to 17 household members. On average, years of education for adults is 10.81. While the average adults' years of age is 37.85, ranging from 18 to 85. The average age of the household head is 46.62.

Generally, our *t*-test results show that the mean difference between treatment and control households is only statistically significant (at least 5%) for household size and adult years of education (Table 2). This implies that household size is higher for control households (5.31) than treatment households (4.92). Interestingly, we also found that control households (11.48) have high years of adult education than the treatment households (10.58). These two characteristics could influence how households seek or secure financial resources. In other words, it could have an impact on the ability of the households to access food from markets and income levels (i.e., poverty levels) to finance other basic needs at the household level. We also found that the average percentage of household heads that are either married or living together for the control group is 72%, which is significantly higher than 67% for the treatment group (Table 2).

Conversely, we did not find strong evidence on whether the mean differences are statistically different for other demographic characteristics. This suggests that the randomization when recruiting households worked adequately as suggested by the balance test results in Table 2. However, these are only descriptive results where other confounding factors are not considered. Thus, these results should be interpreted with caution in this report.

3.1.2 | Demographic characteristics by research group and district

So far, we have disaggregated our results by research groups (treatment vs. control households). While this is informative, this approach cannot give us demographic characteristics at district level. To address this issue, we also did our analysis at district level. Although it is possible to do this analysis for all variables in our study, we only focused on a few variables as guided by mean *t*-test results. Figure 2 shows average household size, by district and research group.

We found that the average household size for the treatment group is higher in Mangochi, seconded by Mzimba, while Zomba district has the lowest average household sizes among the treatment group. In contrast, the average household size for the control group is also higher in Mangochi which is seconded by Mzimba and Dedza districts. Zomba district also has a lower average household size for the control group among all the sampled districts. However, in general, our *t*-test results on the mean difference between control and treatment groups for the household size in the six districts suggest no statistically significant difference between the two groups except for Kasungu.

Adult years of education by district and research group are reported in Figure 3. On average, adult years of schooling among the treatment group is highest in Mzimba, followed by Kasungu and Dedza districts. Conversely, Mangochi has the lowest average adult years of education for the treatment group among the six districts. We observe a similar pattern for the control group. Mzimba district also has a higher average adult year of education for the control

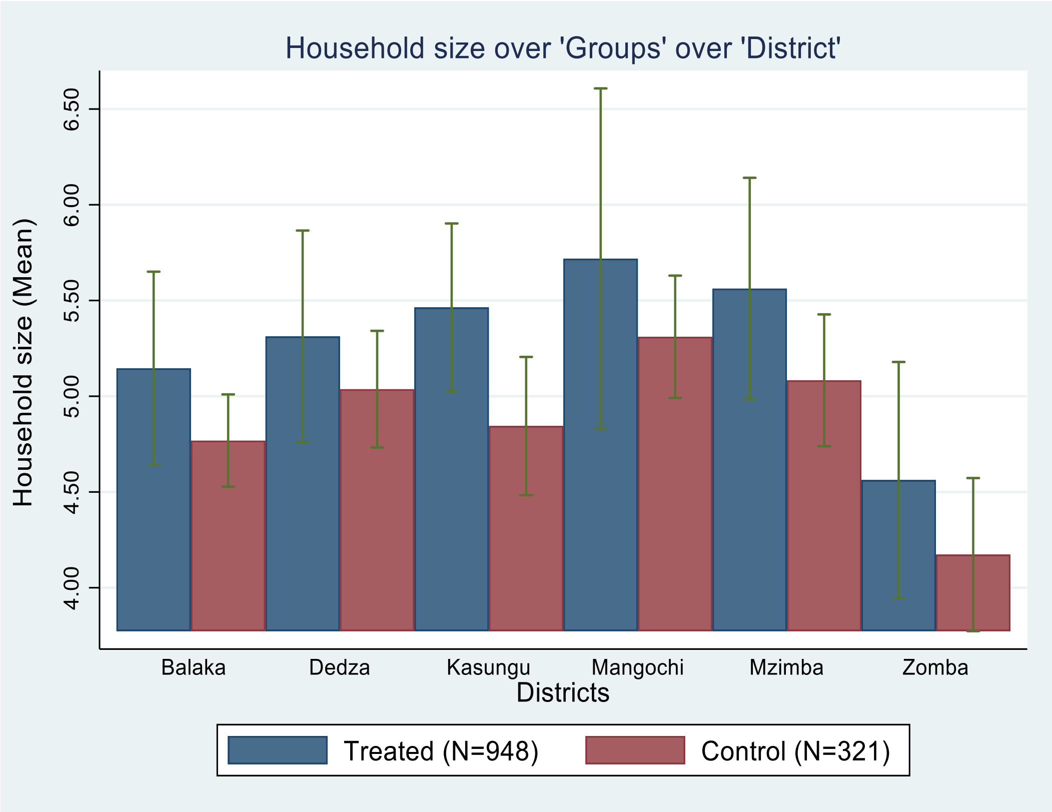


Figure 2: Average household size, by district and research group.

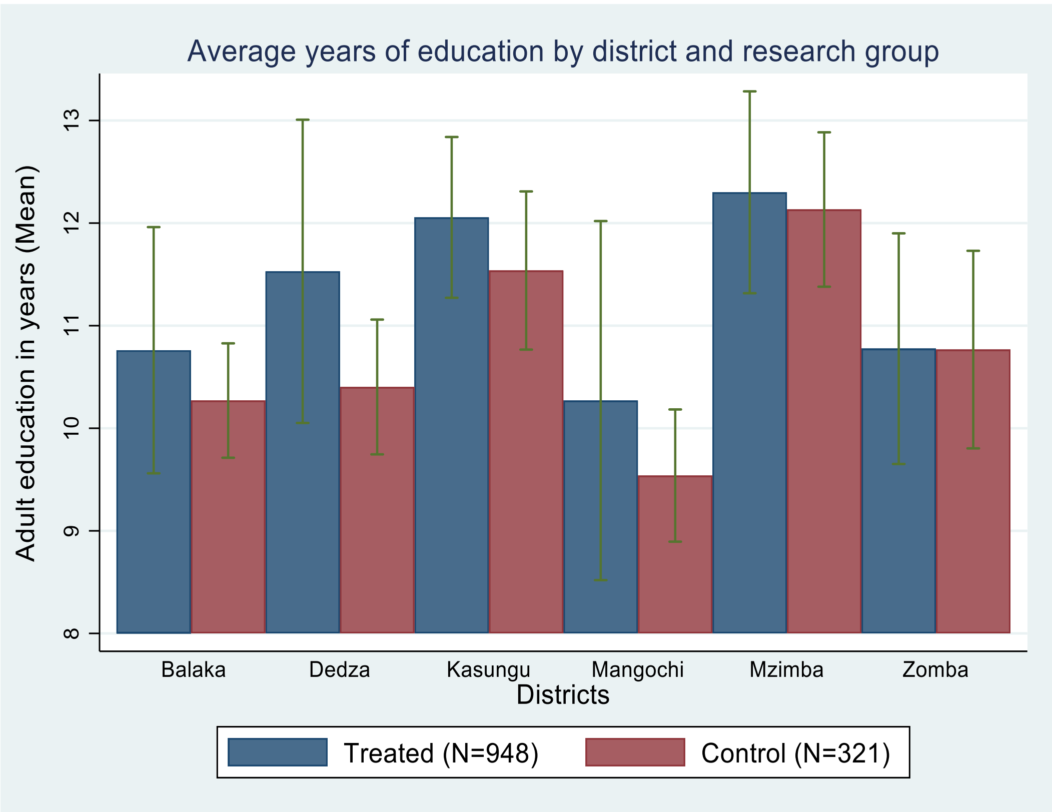


Figure 3: Average adult years of education by district and research group.

group, Kasungu and Dedza follow. Mangochi also has the lowest average adult years of education among the control groups. These results could substantially affect the adoption of SI-MFS interventions. Several existing studies found that education is key for farmers to know the inherent characteristics of agricultural technologies and their benefits (Khonje et al., 2022a; 2018; Manda et al., 2016), particularly if a new agricultural innovation is introduced in an area.

Before concluding the discussion on demographic characteristics, a few other characteristics were analysed and are presented in Table 3. On religion, 77% of the household

Table 3: Summary statistics for other demographic characteristics for the household head by research groups

	Full sample (N=1,269) (1)	Treatment (N=948) (2)	Control (N=321) (3)	Mean difference (4) = (2) – (3)
Religion (%)				
Christianity	77.30	76.27	80.37	-04
Muslim	21.12	22.26	17.76	05*
African traditional religion	1.10	1.16	0.93	00
None	0.47	0.32	0.93	-00
Gender (%)				
Male	66.98	65.93	70.09	-04
Education (%)				
None	13.40	14.14	11.21	03
Standard 1-4	21.59	22.15	19.94	02
Standard 5-8	42.40	41.14	46.11	-05
Form 1-3	12.69	12.97	11.84	01
O-level and above	9.38	9.28	9.66	-00
Adult literacy	0.55	0.32	1.25	-01*

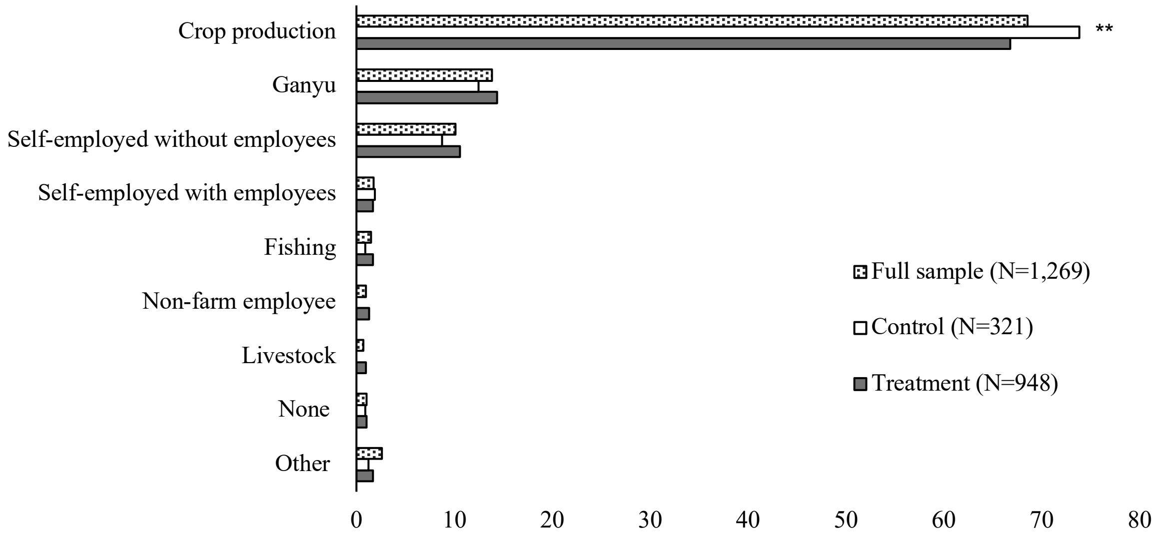
Note: N, Number of observations (No. of farming households). *Significance at 10%.

heads are Christians, while 21% are Muslims, and 1% belong to the African traditional religion. Table 3 also shows that 67% of the households are male headed, with the control group registering a higher proportion (70% vs. 66%) than the treatment group.

On educational background, most household heads (87%) have attended formal education. For those that have attained formal education, 22% of the household heads have attended primary school from standard 1 to 4, and a bigger proportion (42%) have attended primary school from standard 5 to 8. Instead, only 13% of the sampled households had attended secondary education from Forms 1 to 3, and only 9% had attained O-level (Form 4) and beyond. We also found that very few households had attended adult literacy education in the six sampled districts. Overall, after *t*-test analysis on the mean difference between control and treatment groups, statistically significant results were observed for Muslims and farmers who attended adult literacy education.

3.2 | Economic activities

To support their livelihoods, the sampled farming households are involved in several economic activities. Figure 4 reports selected main economic activities. On average, of the 1,269 households, 69% were engaged in crop production as their primary economic activity. While 14% of the sampled households depend on *ganyu* (casual labor) as their main activity, and 10%



Note: N, Number of observations (No. of farming households). **Significance at 5%.

Figure 4: Economic activities by research groups.

are self-employed but without employees as their primary activity (Figure 4). Unlike for *ganyu* and self-employed without employees, the proportion of households involved in crop production was higher in the control group as compared to the treatment group.

How do farming households allocate their time on main economic activities? To provide insights on this question, the research group's time allocation on main economic activities was analysed. Results are reported in Table 4. As expected, most (81%) of the sampled households spend more time on farming during the rainy season, with the treatment group registering a higher proportion (2.64% vs. 0.62%) than the control group. The second-ranked activity during the rainy season was domestic work at 35% and non-farm income generating activities at 31%. These activities were ranked second in the rainy season.

In contrast, most sampled households also spent most of their time on farming followed by non-farm income-generating activities in the dry season. This is the time; farming households are engaged in winter cropping or *dambo* farming or irrigation farming. In addition to winter farming and non-farm income generating activities, the sampled households were fairly engaged in unpaid domestic work (25%) and resting (24%).

Table 4: Summary statistics of time allocation and perceived engagement in daily activities (%)

	Full sample (N=1,269) (1)	Treatment (N=948) (2)	Control (N=321) (3)	Mean difference (N=1,269) (4) = (2) – (3)
Activities ranked first in the rainy season				
Domestic work	2.13	2.64	0.62	02**
Farming	90.78	90.30	92.21	-02
Non-farm income generating activities	5.91	6.01	5.61	00
Socialization	0.32	0.21	0.62	-00
Resting	0.87	0.84	0.93	-00
Activities ranked second in the rainy season				
Domestic work	34.94	34.75	35.51	-01
Farming	6.09	5.93	6.54	-01
Non-farm income generating activities	30.67	31.67	27.73	04
Socialization	10.20	10.59	9.03	02
Resting	18.10	17.06	21.18	-04*
Activities ranked first in the dry season				
Domestic work	17.89	17.62	18.69	-01
Farming	42.16	41.35	44.55	-03
Non-farm income generating activities	34.44	35.23	32.09	03
Socialization	1.81	1.90	1.56	00
Resting	3.70	3.90	3.12	01
Activities ranked second in the dry season				
Domestic work	24.88	25.27	23.75	01
Farming	21.30	21.86	19.69	02
Non-farm income generating activities	13.99	13.97	14.06	-00
Socialization	15.74	15.88	15.31	00
Resting	24.09	23.03	27.19	-04
Average hours spent sleeping during the rainy season	8.21	8.19	8.28	01
Average hours spent sleeping during the dry season	9.01	8.98	9.10	02

Note: N, Number of observations (No. of farming households). *Significance at 10%, **Significance at 5%.

3.3 | Agricultural land

Table 5 shows the summary statistics of household level agricultural land characteristics for the treatment and control groups. The SI-MFS initiative collected data on 2,233 parcels and 2,733 plots of land across 1,269 households. The average land holding size was 0.98 hectares, with the control group having more (1.10 vs. 0.94) agricultural land than the treatment group.

One potential implication of this finding is that the SI-MFS initiative could target promoting technologies that maximize yields per unit area. This is so because most households in the target districts for the initiative have smaller land holdings, and arable land is a major constraint due to high population growth (Rao et al., 2019; Doss et al., 2020; Ricker-Gilbert et al., 2014; United Nations, 2015).

After doing a *t*-test analysis, we found that the average land holding size is higher (1.10 vs. 0.94) and statistically significant for the control group than the treatment group (Table 5). However, we also found that per capita land holding size for the study population is 0.23. The mean difference for the two groups is almost the same and is not statistically significant. On average, we also found that the control group has a significantly higher number of parcels of

Table 5: Summary statistics of agricultural land by research group

	N	Mean	SD	Min	Max
Farming land size (ha)	1,269	0.98	1.45	0.05	39.05
Treatment	948	0.94*	1.53	0.10	39.05
Control	321	1.10	1.17	0.05	12.95
Per capital land size (ha)	1,269	0.23	0.37	0.01	7.81
Treatment	948	0.23	0.40	0.01	7.81
Control	321	0.23	0.26	0.02	2.59
Number of parcels per household	1,269	1.76	0.86	1	8
Treatment	948	1.74*	0.85	1	8
Control	321	1.83	0.91	1	6
Percentage of HH with closest parcels (<15 minutes)	1,269	50	50	1	100
Treatment	948	50	50	1	100
Control	321	50	50	1	100
Percentage of HH with closest parcels (15 to 30 minutes)	1,269	21	41	1	100
Treatment	948	21	41	1	100
Control	321	21	41	1	100
Percentage of HH with closest parcels (30 to 60 minutes)	1,269	18	38	1	100
Treatment	948	17	37	1	100
Control	321	21	40	1	100
Percentage of HH with closest parcels (>60 minutes)	1,269	11	31	1	100
Treatment	948	12*	32	1	100
Control	321	08	27	1	100
Number of plots per household	1,269	2.15	1.15	1	8
Treatment	948	2.14	1.16	1	8
Control	321	2.19	1.14	1	7

Note: N, Number of observations (No. of farming households). SD, standard deviation. HH, Household head. * Denotes the mean difference between treatment group and control group is significant at 10%.

1.83 than that of the treatment group with 1.74 parcels per household. Similarly, the control group has a significantly higher number of plots of 2.19 than that of the treatment group with 2.14 plots per household. We also found that the treatment group had more parcels which were far (>60 minutes) compared to the control group.

Although the average land holding size by research group is informative, this does not give us comprehensive insights at the district level. Does the average land holding size significantly vary in the six study districts? Figures 5 to 7 attempt to provide an answer to this question.

Figures 5, 6, and 7 summarize agricultural land variables disaggregated by district and research group. On average, farming households in Kasungu have relatively bigger land-holding sizes than those in other districts (Figure 5). This is followed by those in Mzimba and Balaka. Farmers in Dedza and Mangochi have the smallest land holding size for both the treatment and control groups.

However, the story is different if we consider number of parcels owned by the farming households. Figures 6 and 7 suggest that the average number of parcels owned by households is highest for the treatment group in Zomba. Mzimba and Mangochi follow this.

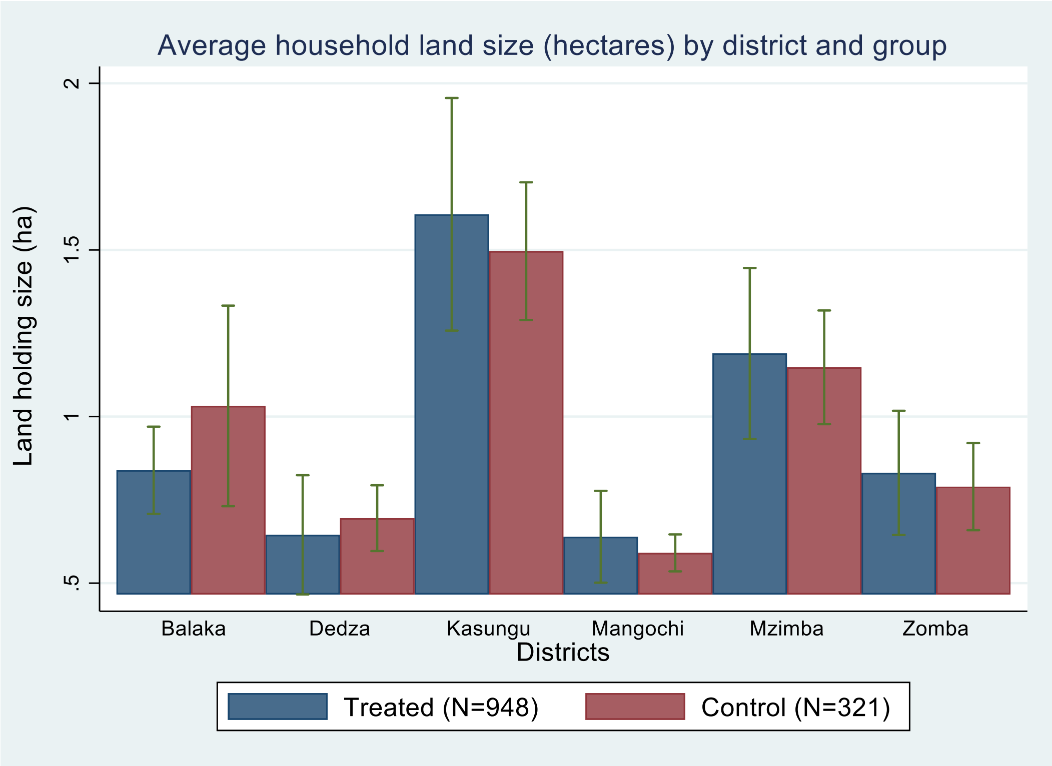


Figure 5: Average household land size (ha) by district and research group.

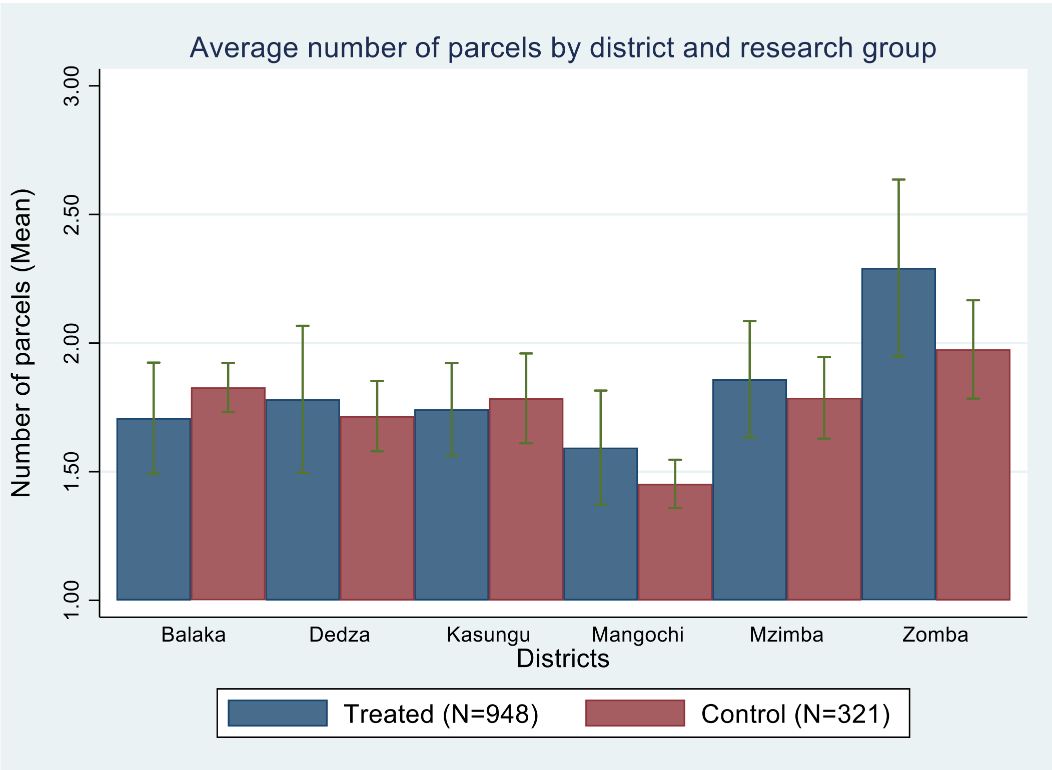


Figure 6: Average number of parcels by district and research group.

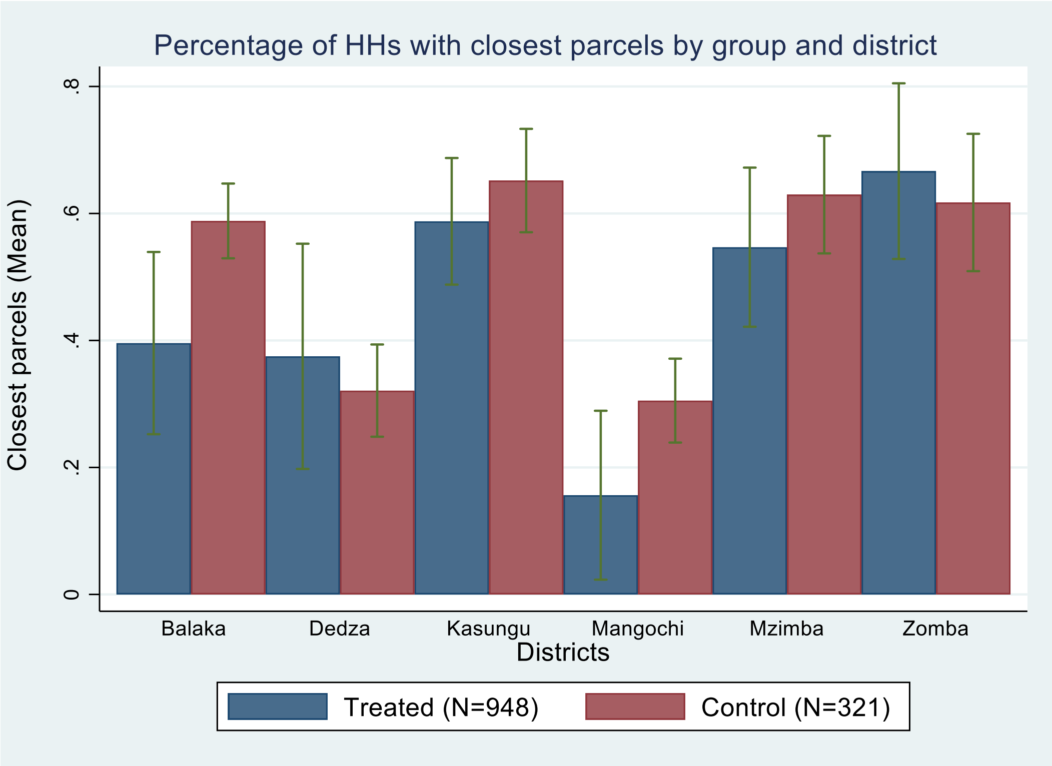


Figure 7: Percentage (%) of households with closest parcel within 15 minutes away by district and research group.

Table 6: Descriptive statistics on parcel characteristics by gender

	Full sample (N=6,368)	Husband		Wife	
		Yes (N=791)	No (N=5,577)	Yes (N=734)	No (N=5,634)
	(1)	(2)	(3)	(4)	(5)
Parcel characteristics					
Distance from homestead to parcel (walking minutes)	21.52	22.60*	20.41	20.74	22.47
	(30.68)	(32.97)	(28.09)	(28.47)	(33.17)
Parcel size (ha)	0.90	0.98**	0.82	0.80***	1.04
	(1.83)	(1.89)	(1.76)	(1.72)	(1.95)
Plot allocation decision (%)	82	100***	63	80***	84
	(39)	(00)	(48)	(40)	(36)
Number of plots (number)	1.30	1.31	1.29	1.27**	1.34
	(0.62)	(0.64)	(0.60)	(0.59)	(0.66)

Note: Mean values are shown with standard deviations in parentheses. We used *t*-test to test the mean difference between columns (2) or (4) and (3) or (5). *Significance at 10%, **Significance at 5%, ***Significance at 1%.

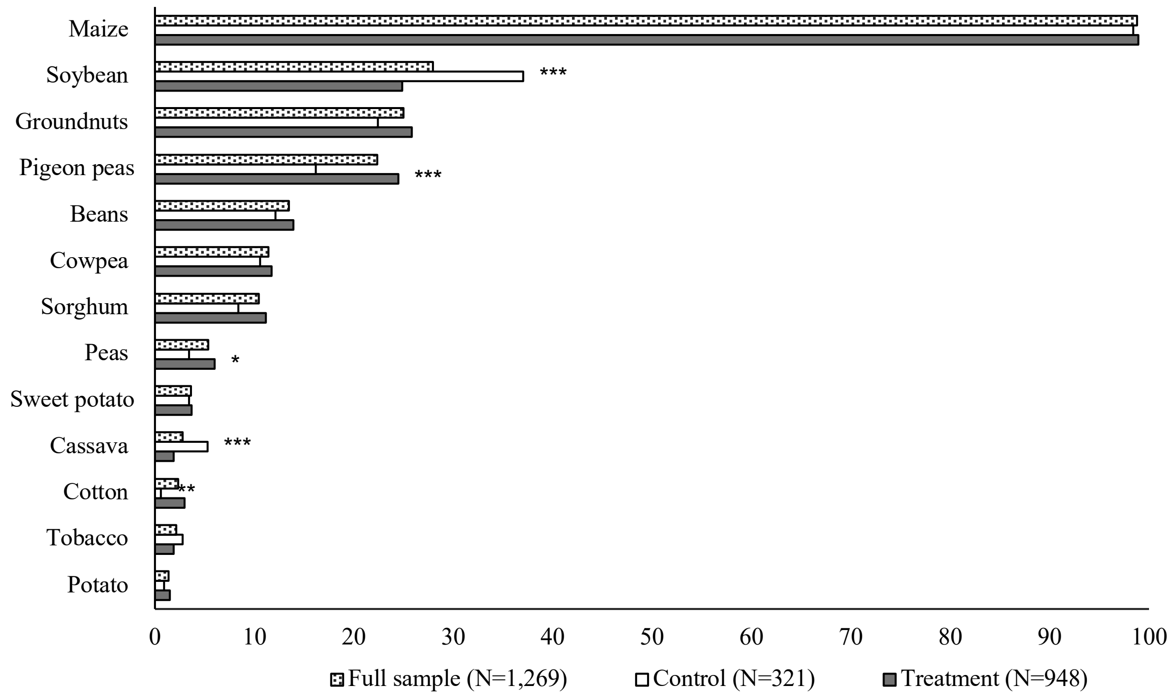
We also analysed parcel-level characteristics by gender (i.e., household decisions by husband vs. wife separately). The results are shown in Table 6. On average, wives owned less farming land (0.80 vs. 0.98) as well as number of plots than husbands (Table 6). We also found that plot allocation decisions at parcel level is primarily done by husbands than wives. Even though, these results are only descriptive, it is worrisome that women are owing less productive assets such as land. Perhaps, this could be an incentive for the SI-MFS initiative to have interventions that are sensitive to gender equality and social inclusion.

3.4 | Agricultural production, inputs and productivity

This section outlines key crop grown, input usage, crop yields and different agricultural practices used by the sampled farmers.

3.4.1 | Crops grown in study districts

Figure 8 reports the proportion of sampled households who grew selected crops by the research group. One key message from Figure 8 is that maize arguably remains the most essential crop



Note: N, Number of observations (No. of farming households). *Significance at 10%, **Significance at 5%, ***Significance at 1%.

Figure 8: Percentage of households who cultivate various crops.

grown on average by 99% of the sampled farmers followed by legume crops, especially soybean, groundnuts, and pigeon peas.

Another key feature from Figure 8 is that soybean and cassava were grown more by farmers in the control group than those from the treatment group. In contrast, almost all legume crops (except soybean) were grown more by farmers in the treatment group than the control group. Probably, this could be a good starting point to promote maize-legume based technologies under the SI-MFS initiative.

Do key crops in Figure 8 also get a fair share of land allocation in a growing season? Table 7 shows the average amount of land allocated to various crops by the sampled farming households. Consistent with Figure 8 results, we also found that main crops such as maize, soybean, and groundnut were allocated more land than other crops.

Table 7: Cultivated crop area (in hectares) by research group

	Full sample (N=1,269) (1)	Treatment (N=948) (2)	Control (N=321) (3)	Mean difference (4) = (2) – (3)
Maize	0.48	0.47	0.52	-0.04*
Soybean	0.13	0.11	0.20	-0.09***
Groundnut	0.08	0.08	0.08	0.01
Pigeon pea	0.03	0.03	0.02	0.01
Tobacco	0.02	0.01	0.02	-0.01
Sorghum	0.02	0.02	0.01	0.00
Cowpea	0.01	0.02	0.01	0.00
Cotton	0.01	0.02	0.00	0.01**
Sweet potato	0.01	0.01	0.01	0.00
Peas	0.01	0.01	0.00	0.00
Cassava	0.01	0.01	0.01	0.01
Bean	0.00	0.00	0.01	-0.00
Potato	0.00	0.00	0.00	-0.00

Note: N, Number of observations (No. of farming households). *Significance at 10%, **Significance at 5%, ***Significance at 1%.

Conversely, farmers allocated small land to minor crops such as tuber crops. This is also suggesting that the SI-MFS initiative could target to promote technologies associated with minor crops such as cassava. Since some of these minor crops are drought tolerant, and promoting them could be critical during extreme weather events.

3.4.2 | Sustainable intensification of mixed farming systems (SI-MFS) technologies

We now focus our discussion on the adoption of SI-MFS technologies by the sampled farmers. Table 8 shows the proportion of sampled farmers who adopted different SI-MFS technologies by research group.

As part of MFS (i.e., combination of crop production and livestock or manure), crop production was done by all sampled farmers. While, livestock farming and organic manure was adopted by 35% of the sampled farmers (Table 8). Generally, mixed farming is more popular (37 vs 31) among the treatment group as compared to the control group (Table 8).

On specific SI-MFS technologies, as expected, our results suggest that inorganic fertilizer application (75%), crop rotation (53%), organic manure (49%) and intercropping (49%) are among critical and popular SI practices adopted by farmers (Table 8). Use of inoculants and agricultural lime is not popular among farmers. The key question is why are such technologies are still unpopular with farmers? Can the SI-MFS initiative take this as an opportunity to enhance the promotion of inoculants in the study districts, especially for districts where legume crops are mainly produced? This could probably help to increase yields for legume crops such as soybean. Of late, soybean is becoming a key strategic cash crop for Malawian farmers, following declining output prices for other key cash crops such as tobacco.

Table 8: Proportion of sampled farmers adopting SI-MFS technologies by research group

	Full sample (N=1,269)	Treatment (N=948)	Control (N=321)	Mean difference
	(1)	(2)	(3)	(4) = (2) – (3)
Mixed farming	35	37	31	06**
	(48)	(48)	(46)	
Fertilizer use	75	75	77	-02
	(43)	(43)	(42)	
Maize improved	44	42	50	-08**
	(50)	(49)	(50)	
Intercropping	49	54	35	19***
	(50)	(50)	(48)	
Crop rotation	53	51	61	-10***
	(50)	(50)	(49)	
Zero/minimum tillage	15	14	16	-01
	(35)	(35)	(36)	
Organic manure	49	51	43	08**
	(50)	(50)	(50)	
Inoculant	02	02	03	-00
	(15)	(15)	(17)	
Lime	01	01	00	01*
	(09)	(11)	(00)	

Note: Mean values are shown with standard deviations in parentheses. N, Number of observations (No. of farming households). *Significance at 10%, **Significance at 5%, ***Significance at 1%.

However, we also found that more farmers are using inorganic fertilizers maybe because they are aware of the benefits associated with fertilizer application on yield gains. One important issue to be considered is that Malawi has been subsidizing fertilizer, improved cereal and legume seeds since the 2004/2005 growing season (Khonje et al., 2022a; 2022b). Evidence suggest that the input subsidy program (ISP) has increased fertilizer usage and adoption of improved maize varieties and to some extent crop yields (Abman & Carney, 2020; Chirwa & Dorward, 2013; Ricker-Gilbert & Jayne, 2017).

Therefore, it is unsurprising to observe that per capita fertilizer use was slightly higher, averaging 111 kg per household (Table 9), with Mzimba and Kasungu, registering highest values of 168 and 155, respectively (Figure 10). Similarly, implementation of the ISP, probably explains why use of improved maize varieties was relatively high (44%), with Mzimba and Mangochi registering adoption rates of 56% and 54%, respectively (Figure 11). As previously explained, this is consistent with existing studies (e.g., Abman & Carney, 2020; Chirwa & Dorward, 2013; Ricker-Gilbert & Jayne, 2017). There was also a notable difference in adoption of improved maize varieties in districts such as Balaka and Dedza between the treatment group and control groups.

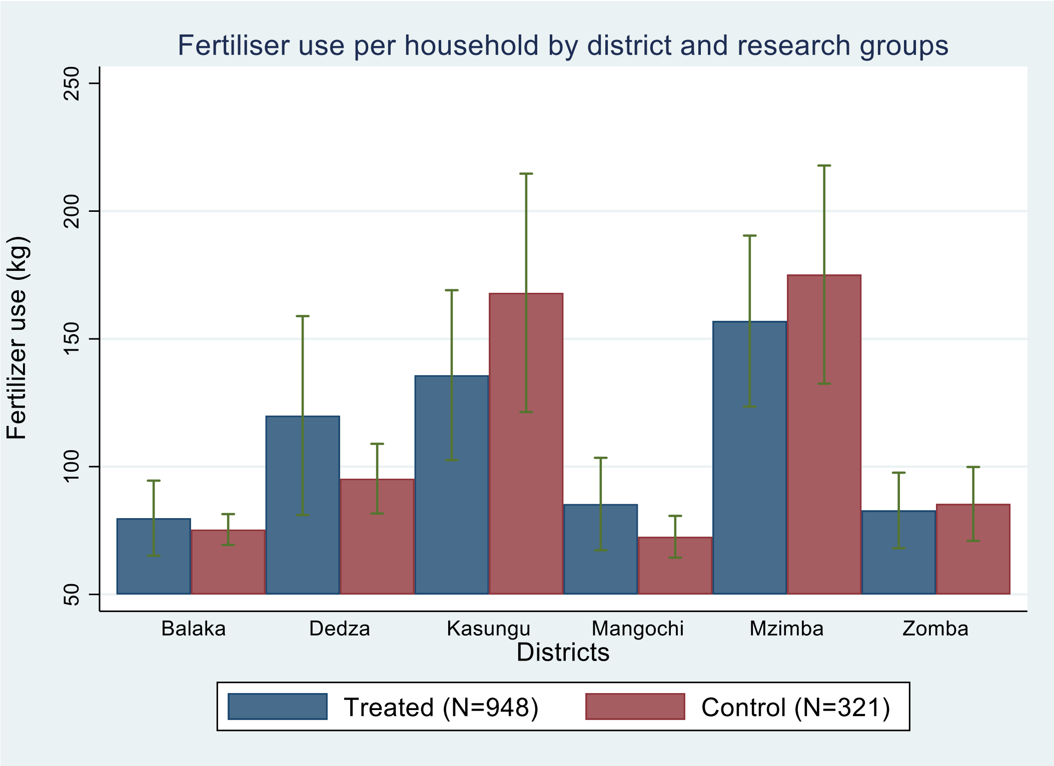


Figure 10: Average amount of fertilizer use by district and research group.

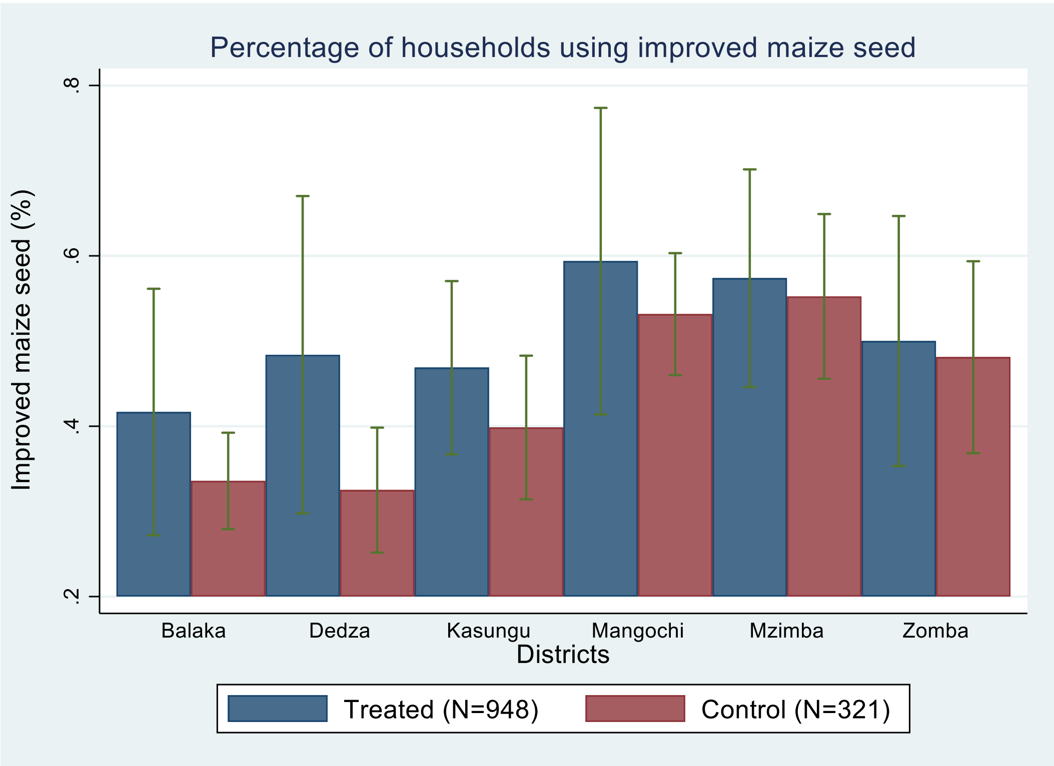


Figure 11: Percentage of households using improved maize seed by district and research group.

3.4.3 | Other agricultural inputs

Apart from using fertilizer and improved maize seed, farmers often use other agricultural inputs. The results on other agricultural inputs are reported in Tables 9, 10 and Figure 12. Only

Table 9: Households using agricultural inputs

	Full sample (N=1,269) (1)	Treatment (N=948) (2)	Control (N=321) (3)	Mean difference (4) = (2) – (3)
Fertilizer use (Kgs)	111.20	108.28	119.63	-11.35
Households using pesticides (%)	10.80	10.86	10.59	0.00
Households using hired labor (%)	21.75	22.47	19.63	0.03
Households using communal labor (%)	42.55	42.09	43.93	-0.05
Personal days (Total person-days used)				
Male	64.09	62.76	67.99	-3.38
Female	75.57	74.81	77.82	-0.58
Communal	9.07	9.02	9.22	-0.35
Male and female	139.66	137.58	145.81	-3.97
Male, female and communal	148.73	146.60	155.03	-3.62
Value of agricultural inputs in MWK				
Fertilizer purchased per household	29,293	27,156	35,608	-8,452**
Improved seeds purchased per household	4,355	4,605	3,728	877
Traditional seeds purchased by household	1,710	1,754	1,571	183

Note: N, Number of observations (No. of farming households). MWK, Malawi Kwacha. **Significance at 5%.

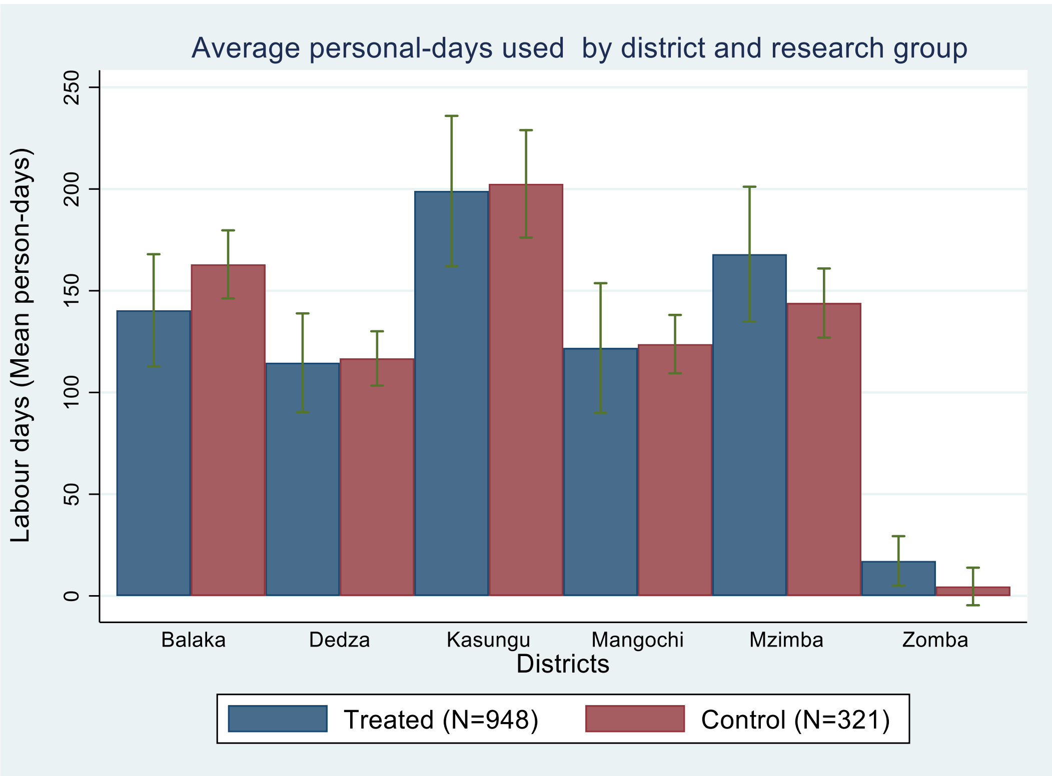


Figure 12: Average person-days used (including communal labor) by district and research group.

Table 10: Descriptive statistics on agricultural inputs by gender

	Full sample (N=6,368) (1)	Husband		Wife	
		Yes (N=791) (2)	No (N=5,577) (3)	Yes (N=734) (4)	No (N=5,634) (5)
Plot size (ha)	0.38	0.41***	0.36	0.37**	0.40
	(0.30)	(0.30)	(0.29)	(0.30)	(0.30)
Improved maize seed (%)	0.33	0.35**	0.31	0.33	0.32
	(0.47)	(0.48)	(0.46)	(0.47)	(0.47)
Fertilizer use (%)	0.49	0.50	0.48	0.51**	0.46
	(0.50)	(0.50)	(0.50)	(0.50)	(0.50)
Pesticide use	0.08	0.10***	0.06	0.06***	0.10
	(0.27)	(0.30)	(0.23)	(0.23)	(0.30)
Hired labor (%)	0.20	0.20	0.19	0.20	0.18
	(0.40)	(0.40)	(0.39)	(0.40)	(0.39)

Note: Mean values are shown with standard deviations in parentheses. We used *t*-test to test the mean difference between columns (2) or (4) and (3) or (5). **Significance at 5%, ***Significance at 1%.

11% of the sampled farmers used pesticides in the management of crop pests (Table 9). We do not find strong evidence on whether the use of pesticides is different between the treatment and control groups. However, strong evidence is found once the results are disaggregated by gender (Table 10). The decision on input use including pesticides, fertilizer, and improved maize seed is primarily done by husbands as opposed to wives. The results further show that wives owned less land than husbands and the implication of this finding has already been discussed in the previous section (i.e., section 3.3).

On farm labor use, both communal labor (43%) and hired labor (22%) were relatively common among the sampled farmers. Using labor days to capture time use and labor on agriculture activities, we also observe that the total amount of family labor plus communal labor used per household was 148.73 personal days on average. We also disaggregated labor use data by gender. On average, the male family labor was reported to be 64.09, which is lower than the female family labor, which was reported at 75.57 personal days. Kasungu and Mzimba registered higher person days on labor use (Figure 12). In contrast, Zomba ranked the lowest on labor use among the six study districts.

3.4.4 | Crop yields

Table 11 reports descriptive statistics for crop yields (kg/ha) by research group. On average, we found that sweet potatoes had highest yield of 2,073. This was followed by irish potatoes (2,000) and cassava (1,701). Based on Figure 8, main crops such as maize, soybean and groundnut had lower yields (kg/ha), on average, ranging from 793 to 1,283. Even though some crops are widely grown in Malawi, their yields are relatively low. Several existing studies suggest that low adoption of agricultural technologies, especially fertilizer and improved seeds,

Table 11: Descriptive statistics for crop yields (kg/ha) by research group

Crop yield (kg/ha)	Full sample (N=1,269) (1)	Treatment (N=948) (2)	Control (N=321) (3)	Mean difference (4) = (2) – (3)
Bean	580.59	611.53	485.38	126.15
	(615.35)	(655.54)	(465.09)	
Groundnuts	793.25	816.82	713.05	103.77
	(824.83)	(834.82)	(790.27)	
Soybean	820.65	850.43	761.85	88.59
	(609.86)	(656.88)	(501.72)	
Cotton	851.86	846.55	926.25	-79.70
	(607.86)	(590.50)	(1135.26)	
Peas	853.82	688.76	1679.10	-990.34***
	(1028.74)	(790.16)	(1615.08)	
Pigeon pea	482.17	464.96	667.19	-202.24
	(394.65)	(372.12)	(632.18)	
Cowpea	790.87	774.53	848.91	-74.38
	(653.95)	(667.07)	(612.64)	
Maize	1283.29	1279.82	1293.61	-13.80
	(932.96)	(948.34)	(887.12)	
Sorghum	1095.00	1022.68	1376.23	-353.55
	(1014.63)	(963.67)	(1170.07)	
Cassava	1700.77	1711.62	1689.29	22.33
	(1425.42)	(1378.72)	(1515.80)	
Tobacco	1398.52	1576.65	1042.26	534.39*
	(793.04)	(760.74)	(774.60)	
Potato	1999.54	1947.77	2241.11	-293.34
	(1295.66)	(1271.32)	(1678.66)	
Sweet potato	2073.01	1975.76	2382.43	-406.67
	(1231.43)	(1211.78)	(1301.12)	

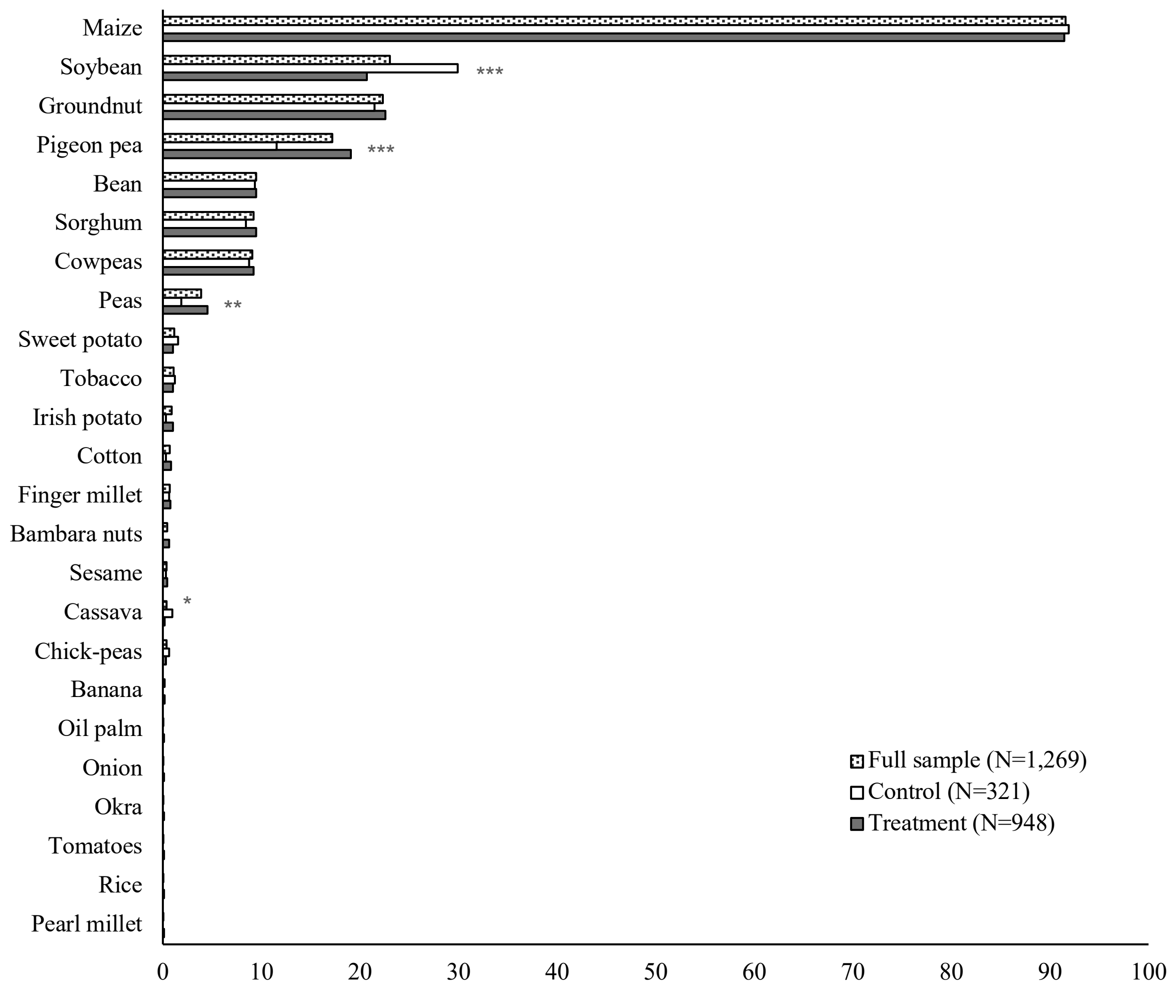
Note: Mean values are shown with standard deviations in parentheses. N, Number of observations (No. of farming households). *Significance at 10%, ***Significance at 1%.

explain why crop yields including cereals and legumes, have remained low (<3.5 tons/ha) in Malawi (Benson, 2021; FAO, 2021; Khonje et al., 2022a; 2022b; 2018; Manda et al., 2016).

The disaggregated results suggest that the control group had higher yields for peas than the treatment group (Table 11). In contrast, the treatment group registered higher yield for tobacco than the control group. Nevertheless, these results should be interpreted cautiously as measurement errors are common when measuring crop yields in agricultural surveys, particularly if recall data methods are used (Abay et al., 2021; 2023; Wollburg et al., 2021). Moreover, more often most farmers rarely keep records on these characteristics, at least in an African context (Khonje et al., 2022a; 2022b).

3.5 | Crop storage

Figure 13 highlights the percentage of households that had various crops in storage a month after harvesting. We found that most (92%) of the sampled households kept maize one month after harvesting. Generally, next to maize, legume crops including soybean (23%), groundnut (22%), pigeon peas (17%) and common beans (9%) were second important crops, which are often stored immediately (i.e., one month) after harvesting.



Note: N, Number of observations (No. of farming households). *Significance at 10%, **Significance at 5%, ***Significance at 1%.

Figure 13: Percentage of households that had crops in storage one month after harvest by research group.

However, conventional wisdom and emerging experimental evidence suggest that more often farmers do not keep legume crops for a longer period because of two post-harvest constraints: (i) lack of appropriate storage technology, and (ii) commitment issues (Nindi et al., 2023). This study (Nindi et al., 2023) suggests that combining technology (i.e., hermetic bags) with collective action (i.e., village storage programs) that is localized and flexible can lead to better post-harvest outcomes for smallholder farmers. For instance, farmers who were offered hermetic bags under the village storage program had multiple benefits from the intervention including, having more legumes at harvest, storing legumes for a longer period, receiving a higher output price, and more revenue than the control group (Nindi et al., 2023).

We also found that, there is a big variation between the two groups on their ability to store crops. For instance, more farmers stored pigeon peas and peas from the treatment group than those from the control group (Figure 13). Conversely, more farmers from the control group stored soybean and cassava. Probably, liquidity constraints and post-harvest constraints explain this behaviour as established by Nindi et al. (2023).

Our sampled farmers were also asked to report on the typed of storage facilities they use for various crops. According to Figure 13, the top three crops that are often kept by farmers are maize, groundnuts, and soybean. Thus, our analysis on type of storage facilities focused on the three crops. The results are reported in Table 12.

Table 12. The proportion of households using various storage facilities

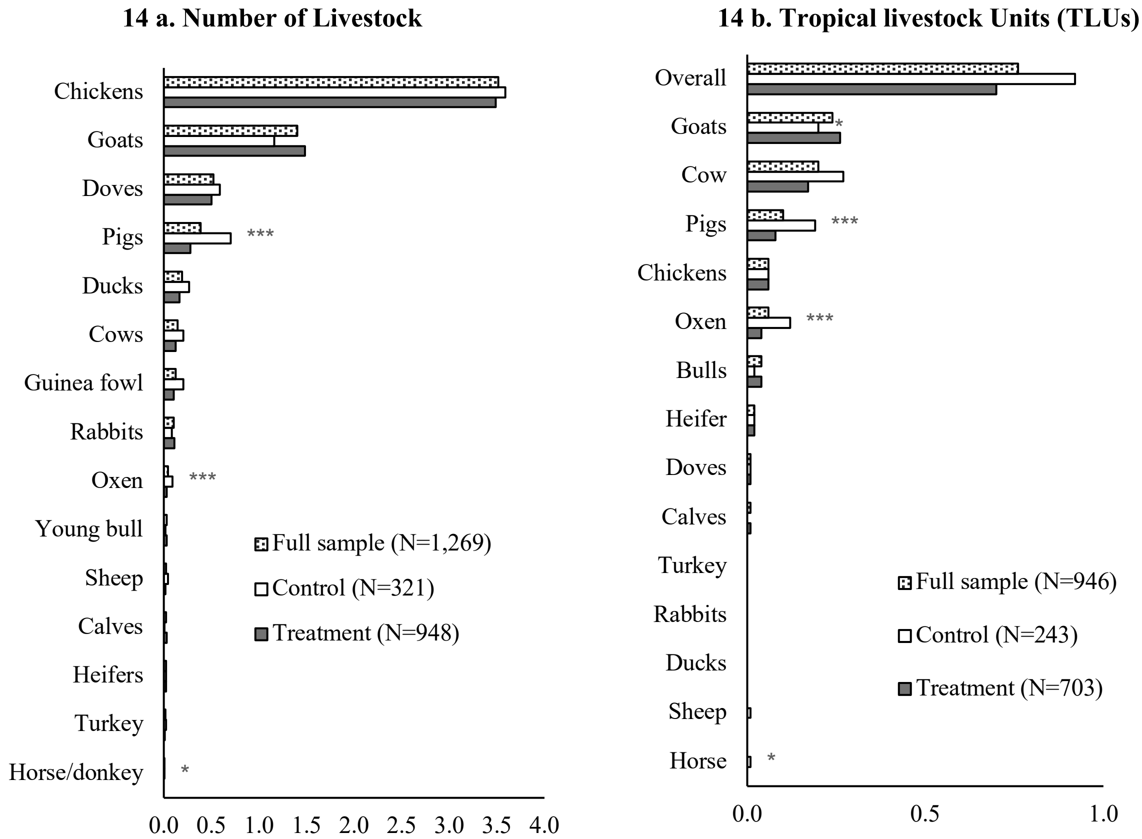
Crop	Storage facility	Full sample (N=1,269)	Treatment (N=948)	Control (N=321)
All crops	Traditional granary	2.52	2.64	2.18
	Metal/plastic silos	1.42	1.48	1.25
	Polypropylene/wollen bags	92.67	92.19	94.08
	Purdue improved crop storage (PICS) bags	3.94	3.80	4.36
	Other	2.99	2.85	3.43
Maize	Traditional granary	2.75	2.88	2.37
	Metal/plastic silos	1.55	1.61	1.36
	Polypropylene/wollen bags	97.07	96.77	97.97
	Purdue improved crop storage (PICS) bags	4.30	4.15	4.75
	Other	2.93	2.65	3.73
Groundnuts	Traditional granary	1.77	2.34	0.00
	Metal/plastic silos	1.77	2.34	0.00
	Polypropylene/wollen bags	99.65	99.53	100.00
	Purdue improved crop storage (PICS) bags	5.65	5.61	5.80
	Other	3.18	2.34	5.80
Soybean	Traditional granary	2.40	2.04	3.13
	Metal/plastic silos	1.71	0.51**	4.17
	Polypropylene/wollen bags	100.00	100.00	100.00
	Purdue improved crop storage (PICS) bags	5.48	4.08	8.33
	Other	3.77	3.06	5.21

Note: N, Number of observations (No. of farming households). ** Denotes the mean difference between treatment group and control group is significant at 5%.

In general, the results show that most farmers (93%) used polypropylene/wollen bags as a storage facility. Specifically, 97% of the farming households used polypropylene/wollen bags as a storage facility for maize, whereas 4.30% of households used other storage facilities. For groundnut and soybean, 99.65% and 100% of the sampled households used wollen bags, respectively. Another interesting feature is that hermetic bags (PICS) are slowly becoming popular next after wollen bags. This suggest that gradual adoption of hermetic bags could help to address post-harvest constraints among smallholder farmers. However, for this technology to be more effective, probably the price differential between wollen bags and hermetic bags should be relatively smaller.

3.6 | Livestock ownership

Figure 14 presents the average number of animals owned by sampled households. Two indicators were used: number of animals and tropical livestock units (TLUs), which are reported in Figures 14a and 14b, respectively. Following prior studies (e.g., Ahmed & Mesfin, 2017; Bellemare & Barrett, 2006; Njuki et al., 2011; Storck et al., 1991), the following TLU conversion factors were used: calf (0.25), heifer (0.75), cow or oxen (1), chicken and other poultry (0.013), sheep and goat (0.13), pigs (0.2), horse (1.10), and rabbits (0.02). Most of the sampled households in the study districts own chickens, goats, and doves. For example, on average, a household kept 3.51 chickens, 1.4 goats, and 0.52 doves (Figure 14a). Overall, a farming household owned 0.76 TLUs (Figure 14b).



Note: N, Number of observations (No. of farming households). *Significance at 10%, **Significance at 5%, ***Significance at 1%.

Figure 14: Average number of animals owned by households by research group.

We get mixed evidence after disaggregating the results by the research group. Figure 14 suggest that the treatment groups had a higher number of young bulls, heifers, sheep, goats and doves than their counterparts. Conversely, the control group had a higher number of ducks, guinea fowl, and turkeys, on average. Nevertheless, the mean differences are statistically significant only for three types of animals: pigs, oxen and horse or donkey (Figures 14a and

14b). As stated earlier, an increase in number of animals per adults would enhance labor productivity and improve food security and health through consumption of ASFs. In addition, livestock ownership is very crucial in providing manure, and livestock income. These resources could be crucial inputs in the SI-MFS initiative. Putting it differently, promoting livestock under the initiative could play a significant complementary role with crop-based farming systems and household resilience.

3.7 | Housing characteristics

Table 13 provides a summary of the housing characteristics for the research groups and for the pooled sample. On average, 69% of the sampled households used burnt bricks as the main material for the wall of the housing unit, and 29% of households used mud bricks for the wall of the house.

Table 13: Housing, water, sanitation and hygiene characteristics (%)

	Full sample (N=1,269) (1)	Treatment (N=948) (2)	Control (N=321) (3)	Mean difference (4) = (2) – (3)
Material for wall				
Mud/mud brick/clay	28.68	29.75	25.55	0.04
Stone/burned bricks	69.11	67.83*	72.9	-0.05*
Cement/sandcrete block	1.97	2.32	0.93	0.01
Material for floor				
Earth/mud/mud brick	74.70	74.58	75.08	-0.00
Cement/concrete	25.06	25.32	24.30	0.01
Material for roof				
Leaves/raffia/thatch	45.15	44.3	47.66	-0.03
Corrugated metal	50.83	51.58	48.6	0.03
Cement/concrete	2.29	2.32	2.18	0.00
Plastic sheeting	1.58	1.58	1.56	0.00
Source of drinking water				
Piped into yard/plot	0.79	1.05	0.00	0.01*
Communal standpipe	4.33	4.11	4.98	-0.01
Open well in yard/plot	1.02	0.95	1.25	-0.00
Open public well	4.26	4.22	4.36	-0.00
Protected public well	1.26	1.27	1.25	0.00
Borehole	85.19	85.23	85.05	0.00
River/Stream	1.50	1.48	1.56	-0.00
Type of toilet				
Private traditional latrine with wall/roof	53.74	52.85	56.39	-0.04
Shared traditional latrine with wall/roof	16.23	16.56	15.26	0.01
Private traditional latrine without wall/roof	20.57	20.68	20.25	0.00
Shared traditional latrine without wall/roof	7.96	8.12	7.48	0.01
Source of lighting				
Electric lights	2.05	2.32	1.25	0.01
Torch	81.25	82.38	77.88	0.05**
Solar panel	12.92	11.81	16.2	-0.04**
Cooking fuel				
Wood	94.96	95.36	93.77	0.02
Charcoal	4.89	4.43	6.23	-0.02

Note: N, Number of observations (No. of farming households). The analysis focused on major characteristics only and as a result they do not add to 100%. *Significance at 10%, **Significance at 5%.

On floor characteristics, we found that 75% of the sampled household's floors are made of mud and 25% are constructed with cement on average. In terms of the roof, about 51% of households thatched their roofs with corrugated metal. While 45% of the sampled households thatched their houses with leaves or raffia or grass.

On water, sanitation and hygiene characteristics, majority (85%) of the sampled households in the study districts sourced their water from boreholes. Most (54%) of the households used their own (private) traditional latrines with a wall and roof. In terms of lighting and cooking fuel, 81% of households used torches for lighting, and 95% used wood for cooking. The mean differences between the treatment and control groups is statistically significant for stone/burned brick walls, piped water in a yard/plot, and two types of lighting sources: torches and solar lights.

3.8 | Household wealth index

We construct an aggregate wealth index using housing conditions, durable non-agricultural assets, durable agricultural assets, livestock ownership, and land ownership as proposed by Hjelm et al. (2017). We compute the aggregate wealth index through factorial analysis using the principal component factor (PCF) method. Table 14 shows the summary statistics of the households in quintiles. The higher the wealth index for a household, the wealthier are its members on average. The results show that most of the households were in the poorest category.

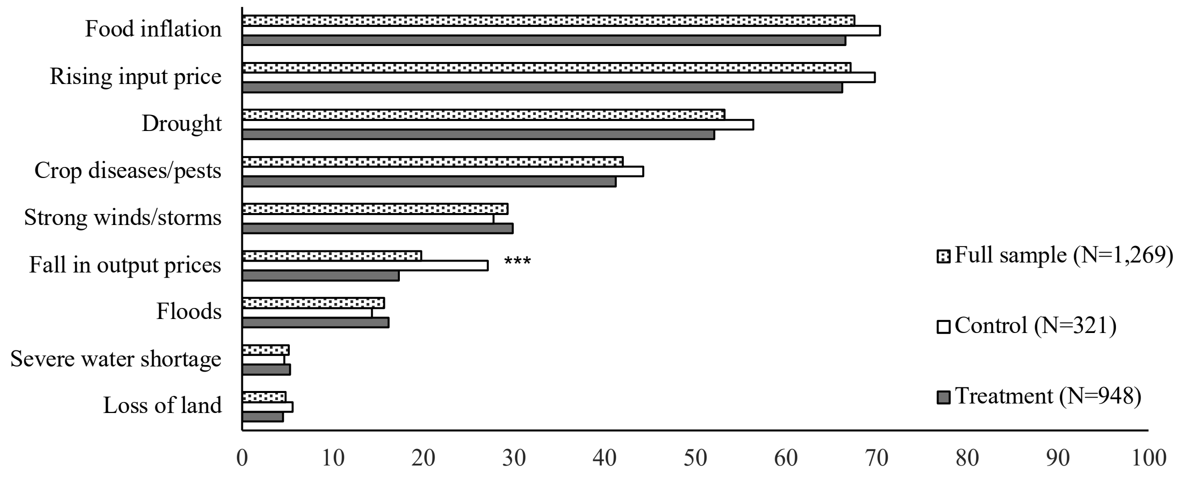
Table 14: Aggregate wealth index

	Number of households	Mean	SD	Min	Max
First quintile (poorest 20%)	238	-1.932	0.256	-2.538	-1.523
Second	237	-1.224	0.174	-1.520	-0.900
Third	237	-0.558	0.210	-0.899	-0.136
Fourth	237	0.426	0.354	-0.134	1.083
Fifth (richest 20%)	237	3.300	2.565	1.086	15.416
Full sample	1,186	0.001	2.165	-2.538	15.416

Note: SD, standard deviation

3.9 | Agriculture-related shocks

Figure 14 provides descriptive statistics on agricultural shocks faced by the farmers over the past five years. On average, we found that rising food prices (68%), rising agricultural input prices (67%), drought (53%) and crop pest and disease outbreaks (42%) as among major agricultural shocks affecting farmers. However, we did not find strong evidence on the mean differences between the two groups for these major agricultural shocks.



Note: N, Number of observations (No. of farming households). ***Significance at 1%.

Figure 15. Percentage of households who experienced agriculture-related shocks.

Conversely, we found that the mean difference between the treatment and control groups was statistically significant for falling agricultural output prices. This suggests that most farmers could not maximize returns from agricultural production.

Perhaps, two important insights emerge from these findings. First, consistent with existing literature (Amadu et al., 2020; MacLaren et al., 2022; Tesfaye et al., 2021), extreme weather events remain one of the formidable challenges facing the current agricultural farming systems for the Global South. Second, current agricultural farming systems should aim at addressing the agricultural shocks in totality if sustainable food production is to be achieved. By working on adaptive research that works for farmers, therefore, this will be the right opportunity for the SI-MFS initiative to contribute several SDGs by 2030.

4 | Results by Impact Areas

As mentioned earlier, the SI-MFS initiative is expected to contribute to five impact areas: (1) nutrition, health and food security; (2) poverty reduction, livelihoods and jobs; (3) gender equality, youth and social inclusion; (4) climate adaptation and mitigation; and (5) environmental health and biodiversity. The five impact areas will contribute to several SDGs including no poverty, zero hunger, good health, gender equality and empowering all women and girls, decent work and economic growth.

To measure the performance and results of the SI-MFS initiative, several indicators will be used for each of the five impact areas. For nutrition, health and food security, we will report three standard indicators: food consumption expenditure, food consumption score (FCS) and household dietary diversity score (HHDS). In addition, crop productivity associated with the initiative, could also be a proxy measure for food security. However, the results on crop yields are not discussed in the proceeding chapter as they are already discussed in chapter three.

While several indicators exist on measuring poverty and welfare, we will focus on three indicators: household income, poverty and proportion of farming households engaged in farm and non-farm activities in this study. These indicators will be key on assessing the performance of the SI-MFS initiative on improving livelihoods of people through (i) higher incomes, and (ii) increasing on-farm and off-farm jobs, which will (iii) lift people out of poverty.

So far, the first two impact areas have direct measurement indicators, which is different with measuring gender equality, youth and social inclusion. In other words, gender analysis cannot be done in isolation. Here, in addition to some of the aforementioned indicators especially on poverty, livelihood, and jobs, we will disaggregate time use and participation in community organization by gender (i.e., male farmers vs. female farmers) and other marginalized groups (youth vs. non-youth). The objective of this analysis is to examine how women, the youth, and other marginalized groups could be empowered.

Lastly, the last two impact areas: climate adaptation and mitigation; and environmental health and biodiversity are somehow interrelated. As a result, the results for the two impact areas are combined. Here, we will report on adoption of climate-smart agricultural innovations promoted under the SI-MFS initiative. For instance, increasing area under SI technologies could be key in improving resilience to climate change for millions of farmers, reducing carbon foot prints and greenhouse gas (GHG) emissions and enhancing environmental health and biodiversity (Holling, 1995; Munandar et al., 2015; Thornton et al., 2018).

4.1 | Nutrition, health and food security

To describe nutrition, health and food security status, we used household-level (a seven-day dietary recall) food consumption data. We calculated three indicators: FCS, HDDS, and food consumption expenditure. FCS is measured by summing a predetermined set of weights—i.e., staples (2), pulses (3), vegetables (1), fruits (1), meat and fish (4), dairy (4), sugar (0.5) and oil (0.5)—designed to reflect the heterogeneous dietary quality for each of the eight food groups consumed by the households (WFP, 2008). The FCS ranges in value from zero to 112, and a higher score would imply a better heterogeneous dietary quality.

HDDS is measured as the count of ten different food groups—i.e., cereals, tubers and roots, legumes and nuts, meat and fish products, vegetables, fruits, milk and milk products, sweets, sugars and beverages, oils and fats, and spices and condiments—consumed by the household during the recall period. These indicators have been widely adopted in measuring household access to nutritious food and food security in the sub-Saharan Africa (Khonje et al., 2022a; Muthini et al., 2020; Sibhatu et al., 2015). The results are reported in Table 15.

Table 15: Descriptive statistics for nutritional outcomes and food consumption expenditure (income) by research groups

	Full sample (N=1,269) (1)	Treatment (N=948) (2)	Control (N=321) (3)	Mean difference (4) = (2) – (3)
Food consumption score	40.82	40.95	40.44	0.52
	(13.54)	(13.54)	(13.55)	
Household dietary diversity score	7.21	7.24	7.12	0.12
	(1.69)	(1.68)	(1.72)	
Food expenditure (MWK/capita/year)	108,981	113,574	95,415	18,158***
	(109,194)	(112,869)	(96,425)	

Note: Mean values are shown with standard deviations in parentheses. 1 US$=MWK1,025 at the time of the survey (i.e., December 2022). ***Significance at 1%.

On average, we found that a farming household had FCS of 41 and HDDS of seven. The treatment group had slightly higher scores than the control group. However, statistically, our *t*-test results suggest that mean difference between the two groups is not substantial (not significant). On the contrary, we found that farming households from the treatment group had higher food expenditure than those in the control group by MWK18,158 (Table 15).

Like most earlier results, we also disaggregated nutritional outcomes and food consumption expenditure (income) by district. The results are reported in Table 16. Farmers in Mangochi had the highest FCS, HHDS and food consumption expenditure than other study districts. While, Dedza had the lowest FCS, HHDS and food consumption expenditure. These results suggest that farmers in Mangochi are able to access more diverse food groups with easy.

Table 16: Descriptive statistics for nutritional outcomes and food consumption expenditure (income) by district

	Balaka (N=320) (1)	Dedza (N=194) (2)	Kasungu (N=232) (3)	Mangochi (N=222) (4)	Mzimba (N=172) (5)	Zomba (N=129) (6)
Food consumption score	38.19	36.55	44.09	46.14	39.68	40.28
	(11.30)	(14.59)	(15.31)	(12.19)	(13.36)	(11.95)
Household dietary diversity score	7.06	6.56	7.47	7.64	7.36	7.18
	(1.51)	(1.96)	(1.76)	(1.36)	(1.83)	(1.52)
Food expenditure	117,806	100,004	101,503	124,177	103,996	94,532
(MWK/capita/year)	(105,877)	(96,993)	(88,845)	(158,377)	(75,506)	(98,794)

Note: Mean values are shown with standard deviations in parentheses. 1 US$=MWK1,025 at the time of the survey (i.e., December 2022).

Table 17: Descriptive statistics for nutritional outcomes and food consumption expenditure (income) by gender and age category

	Gender		Age category	
	Male HH (N=850) (1)	Female HH (N=419) (2)	Youth HH (N=364) (3)	Non-Youth HH (N=905) (4)
Food consumption score	41.82***	38.80	41.13	40.70
	(13.41)	(13.60)	(12.48)	(13.94)
Household dietary diversity score	7.36***	6.91	7.39**	7.14
	(1.65)	(1.72)	(1.65)	(1.70)
Food expenditure (MWK/capita/year)	110,531	105,835	119,326**	104,820
	(104,792)	(117,686)	(94,160)	(114,471)

Note: Mean values are shown with standard deviations in parentheses. 1 US$=MWK1,025 at the time of the survey (i.e., December 2022). **Significance at 5%, ***Significance at 1%.

We also disaggregated nutritional and outcomes and food consumption expenditure by gender and age category. The results are reported in Table 17. Two key story lines are emerging from these results.

First, female headed households have lower scores for FCS and HDDS than male headed households. This finding emphasize that women continue to struggle in accessing resources including income for buying food. This could be related to the fact that women farmers have a lower probability of adopting improved agricultural technologies, mainly because of social norms embedded in wider socio-cultural settings that constrain their access to resources (Gebre et al., 2019; Ndiritu et al., 2014). Thus, the initiative may need to implement gender-sensitive interventions, especially on institutional and systemic barriers on adoption of agricultural technologies at scale.

Second, it is interesting to note that young people have a higher HDDS and food consumption expenditure than adults. However, this may not mean that young people have more resources but rather culturally they are considered as a priority to be given food, in particular during the lean period.

4.2 | Poverty reduction, livelihoods and jobs

In addition to food consumption expenditure data, we also used non-food consumption expenditure data to calculate poverty status. The results are shown in Table 18. Overall, 94% and 74% of the farming households were involved in agriculture and non-agriculture enterprises, respectively as a primary livelihood activity. Majority (73%) of the sampled households are poor[1]. More poor households came from the control group (76%) than the treatment group (72%). Table 18 results also suggest that the treatment group (72%) had few farmers who were involved in non-agriculture activities than the control group (78%).

Table 18: Descriptive statistics on poverty status, income, livelihoods and jobs

	Full sample	Research group		Mean difference
		Treatment	Control	
	(N=1,269)	(N=948)	(N=321)	
	(1)	(2)	(3)	(4) = (2) – (3)
Poor households (%)	73	72	76	04
	(45)	(45)	(43)	
Farmers involved in agriculture (%)	94	93	95	02
	(24)	(25)	(21)	
Farmers involved in non-agriculture activities (%)	74	72	78	06**
	(44)	(45)	(42)	
Household income (MWK/capita/year)	182,204	177,103	197,270	20,167
	(1,102,897)	(1,074,447)	(1184,567)	
Crop income (MWK/capita/year)	166,397	161,416	181,107	19,691
	(1,098,564)	(1,072,904)	(1172,697)	
Livestock income (MWK/capita/year)	185	169	231	62
	(1,757)	(1,740)	(1,806)	
Non-farm income (MWK/capita/year)	15,622	15,518	15,931	413
	(24,225)	(22,613)	(28,496)	

Note: Mean values are shown with standard deviations in parentheses. 1 US$=MWK1,025 at the time of the survey (i.e., December 2022). **Significance at 5%.

Are the poverty levels similar across the study district? To provide insights on this question, we disaggregated poverty indicators by district. The results are reported in Table 19. Using consumption expenditure data, the results show that poverty levels ranged from 69% in Mangochi to 79% in Zomba.

However, the story is different if household—i.e., summing crop income, livestock income and non-farm income—income data is used. For instance, on average, Dedza has the highest household income than any other study district (Table 19). Possibly, farmers in Dedza could have benefited from adoption of SI under the Africa RISING.

[1] We used $1.25 as poverty line and purchasing power parity of 0.39 (https://data.worldbank.org/indicator/PA.NUS.PPPC.RF). The average official exchange rate at the time of the survey was 1 US$=MWK1025 (https://www.exchangerates.org.uk/USD-MWK-spot-exchange-rates-history-2022.html).

Table 19: Descriptive statistics on poverty status, income, livelihoods and jobs by district

	Balaka (N=320) (1)	Dedza (N=194) (2)	Kasungu (N=232) (3)	Mangochi (N=222) (4)	Mzimba (N=172) (5)	Zomba (N=129) (6)
Poor households (%)	72	77	70	69	73	79
	(45)	(42)	(46)	(46)	(44)	(41)
Farmers involved in agriculture (%)	93	97	98	89	95	90
	(25)	(16)	(15)	(32)	(21)	(30)
Farmers involved in non-agriculture activities	73	71	70	80	76	71
(%)	(45)	(46)	(46)	(40)	(43)	(45)
Household income (MWK/capita/year)	84,431	315,819	268,888	112,120	247,016	102,099
	(381,801)	(2145,579)	(833769)	(457,452)	(1,479,874)	(344,162)
Crop income (MWK/capita/year)	71,151	303,751	252,910	91,020	228,610	87,280
	(379,067)	(2,142,655)	(834,063)	(452,639)	(1463,646)	(342,818)
Livestock income (MWK/capita/year)	112	114	305	163	34	495
	(1,271)	(1,266)	(2,496)	(1,473)	(217)	(2,984)
Non-farm income (MWK/capita/year)	13,168	11,955	15,673	20,936	18,372	14,323
	(15,553)	(16,802)	(18,732)	(34,422)	(27,521)	(31,586)

Note: Mean values are shown with standard deviations in parentheses. HH, Household Head. 1 US$=MWK1,025 at the time of the survey (i.e., December 2022).

Table 20: Descriptive statistics on poverty status, income, livelihoods and jobs by gender and age category

	Gender		Age category	
	Male HH (N=850) (1)	Female HH (N=419) (2)	Youth HH (N=364) (3)	Non-Youth HH (N=905) (4)
Poor households (%)	71**	77	65***	76
	(45)	(42)	(48)	(43)
Farmers involved in agriculture (%)	95***	0.91	91***	95
	(21)	(28)	(29)	(21)
Farmers involved in non-agriculture	73	74	62***	78
activities (%)	(44)	(44)	(49)	(41)
Household income (MWK/capita/year)	229,292**	86,679	132,400	202,236
	(1,319,680)	(372,770)	(486,641)	(1,268,727)
Crop income (MWK/capita/year)	213,212**	71,426	115,231	186,976
	(1,314,952)	(368,071)	(486,327)	(1,263,461)
Livestock income (MWK/capita/year)	200	155	34**	246
	(1,921)	(1,366)	(346)	(2,066)
Non-farm income (MWK/capita/year)	15,881	15,098	17,135	15,014
	(23,765)	(25,152)	(24,636)	(24,044)

Note: Mean values are shown with standard deviations in parentheses. HH, Household Head. 1 US$=MWK1,025 at the time of the survey (i.e., December 2022). We used 35 years as cut-off point to define youth (<36 years) and non-youth (>35 years) categories. We used *t*-test to test the mean difference between Male HH and Female HH, and Youth HH and Non-Youth HH. *Significance at 10%, **Significance at 5%, ***Significance at 1%.

Interestingly, we also found that poverty levels were statistically significant among female household heads and the youth compared to male household heads and non-youth (i.e., adults) (Table 20). This could be an incentive for the SI-MFS initiative to significantly contribute towards reduction of poverty in the study districts. However, female headed households and the youth are getting less farm (i.e., crop and livestock) income from agriculture enterprises than their counter-parts (Table 20). Perhaps, these results entail that women farmers and the youths could be deliberately targeted as beneficiaries for the SI-MFS initiative.

4.3 | Gender equality, youth and social inclusion

To provide more insights on gender equality, youth and social inclusion, we also conducted a gender analysis for a few additional indicators. Our analysis focused on agriculture and non-farm activities, time use, and participation in community organizations. The results are reported in Tables 21 and 22.

Table 21: Descriptive statistics on agriculture and non-farm activities by gender

	Full sample (N=6,368) (1)	Husband Yes (N=791) (2)	Husband No (N=5,577) (3)	Wife Yes (N=734) (4)	Wife No (N=5,634) (5)
Panel A: Agricultural activities					
Agricultural activities (%)	94	97***	93	99***	92
	(24)	(17)	(26)	(12)	(26)
Casual labor (%)	48	56***	45	56***	45
	(50)	(50)	(50)	(50)	(50)
Casual labor (MWK/Month)	12,499	35,078***	9,296	15,972	12,046
	(131,094)	(300,496)	(82,145)	(41,398)	(138,565)
Panel B: Non-farm activities					
Non-farm activities (%)	18	31***	15	17	19
	(39)	(46)	(35)	(38)	(39)
Received wages (%)	99	99	99	99	99
	(09)	(11)	(07)	(09)	(09)
Received wages (MWK/Month)	08	24***	06	13**	08
	(55)	(82)	(49)	(68)	(53)
Non-farm business (%)	27	37***	24	30**	26
	(44)	(48)	(43)	(46)	(44)

Note: Mean values are shown with standard deviations in parentheses. 1 US$=MWK1,025 at the time of the survey (i.e., December 2022). We used *t*-test to test the mean difference between columns (2) or (4) and (3) or (5). *Significance at 10%, **Significance at 5%, ***Significance at 1%.

Regarding agricultural activities, the results suggest substantial gender pay gap exist between wives and husband. For instance, male casual laborers earned more income when they were engaged in farming activities than female casual workers (Table 21, Panel A). This is consistent with a few existing studies (e.g., Kim et al., 2023) who found that gender pay gaps in economics and agricultural/applied economics is estimated to be 8.3% and 4.1%, respectively. What is worrisome is that a similar pattern is observed in non-farm activities (Table 21, Panel B). Specifically, male workers earned more than female workers once they were engaged in non-farm activities and businesses. Interventions aimed at reducing this gender pay gap are needed if gender equality is to be achieved.

Could this gender pay gap in both agriculture and non-farm activities be explained by differences in time use? We now report these insights in Table 22. Panels A, B, and C of Table 22 reports time use during dry season, rainy season, and participation in community organizations, respectively. During dry season, male farmers spent most of their time on

Table 22: Descriptive statistics on time use and participation in community organizations by gender

	Full sample (N=6,368) (1)	Husband Yes (N=791) (2)	Husband No (N=5,577) (3)	Wife Yes (N=734) (4)	Wife No (N=5,634) (5)
Panel A: Dry season (%)					
Domestic work	10	07***	11	54***	05
	(31)	(26)	(31)	(50)	(21)
Farming	13	45***	08	34***	10
	(34)	(50)	(28)	(47)	(30)
Non-farm activities	09	41***	04	11***	08
	(28)	(49)	(20)	(32)	(27)
Socialization	00	02***	00	00	00
	(06)	(15)	(04)	(05)	(07)
Resting	01	04***	00	00*	01
	(09)	(20)	(06)	(05)	(09)
Panel B: Rainy season (%)					
Domestic work	02	0.01**	02	10***	01
	(13)	(0.08)	(14)	(30)	(08)
Farming	30	0.91***	21	89***	22
	(46)	(0.29)	(41)	(32)	(41)
Non-farm activities	01	0.07***	01	01	01
	(12)	(0.26)	(08)	(12)	(12)
Socialization	00	0.00***	00	00	00
	(03)	(0.06)	(02)	(00)	(03)
Resting	00	0.01***	00	00	00
	(05)	(0.10)	(03)	(04)	(05)
Panel C: Participation in community organizations (%)					
Input supply	09	11***	07	07*	09
	(28)	(31)	(26)	(26)	(29)
Farmers' cooperative	06	06	05	03***	07
	(23)	(24)	(22)	(18)	(25)
Local administration	06	11***	03	02***	09
	(24)	(31)	(18)	(13)	(28)
Farmers' association	12	13	11	09***	13
	(32)	(33)	(32)	(29)	(34)
Women's association	05	00***	08	10***	02
	(22)	(00)	(27)	(31)	(14)
Church/mosque congregation	44	47**	43	45	44
	(50)	(50)	(49)	(50)	(50)
Village savings group	31	13***	41	45***	23
	(46)	(34)	(49)	(50)	(42)
Government team	04	06***	02	02***	05
	(19)	(24)	(15)	(13)	(21)
Waters user's association	01	01*	01	01	01
	(09)	(11)	(07)	(07)	(10)

Note: Mean values are shown with standard deviations in parentheses. We used *t*-test to test the mean difference between columns (2) or (4) and (3) or (5). *Significance at 10%, **Significance at 5%, ***Significance at 1%.

farming (45%) and non-farm activities (41%). Conversely, female farmers spent most of their time on domestic work (54%) as well as farming (34%).

The story is different in rainy season. As expected, the middle (Panel B of Table 22) panel shows that both male and female farmers increased their time spent on farming to 91% and 89%, respectively. One more interesting feature is that female farmers significantly reduced—i.e., from 54% in dry season to 10% in rainy season—time spent on domestic work between the two periods.

Although both male and female farmers rarely spent more time on socialization, the bottom panel (Panel C of Table 22) show that they do participant in community organizations. We observe two key differences from Table 22. First, on average, male farmers primarily participated in community organizations such as church or mosque (47%), local administration (11%) and groups associated with input supply (11%). Second, female farmers primarily participated in village savings and loan associations (VSLA) (45%) and farmers' associations (9%). These findings imply that interventions aimed at enhancing gender equality and social inclusion could target the aforementioned community organizations.

In general, these results underscore that gender and social inequalities are a persistent feature of agri-food systems including MFS. They relate to deeply entrenched inequitable norms that produce adverse outcomes—primarily for women, the youth, and marginalized actors—and obstruct progress towards the SDGs (McDougall et al., 2020).

4.4 | Climate adaptation, mitigation and environmental health and biodiversity

While climate-smart agricultural innovations promoted under the SI-MFS initiative could be measured by several indicators, we focus on adoption and area under SI technologies, in particular CA (i.e., crop residue incorporation, zero tillage, and crop rotation), inorganic fertilizer, organic manure, and intercropping. The two (i.e., the probability to use SI technologies and area under SI technologies) indicators accounts for both extensive and intensive margins, respectively. Several prior studies (e.g., Khonje et al., 2022a; 2018; Manda et al., 2016; Tesfaye et al., 2021; Zhang et al., 2023), use these technologies as proxy indicators for SI.

Following this literature, the results on percentage of sampled farmers adopting SI technologies by gender are shown in Table 23. Unlike intercropping, CA and use of inorganic

Table 23: Proportion of sampled farmers adopting sustainable intensification technologies by gender

	Full sample (N=1,269)	Male HH (N=850)	Female HH (N=419)	Mean difference
	(1)	(2)	(3)	(4) = (2) – (3)
Conservation agriculture	81	84	75	09***
	(39)	(37)	(43)	
Inorganic fertilizer	75	77	71	06**
	(43)	(42)	(45)	
Organic manure	49	48	50	-01
	(50)	(50)	(50)	
Intercropping	49	46	55	-09***
	(50)	(50)	(50)	

Note: Mean values are shown with standard deviations in parentheses. HH, household head. We used *t*-test to test the mean difference between Male HH and Female HH. N, Number of observations (No. of farming households). **Significance at 5%, ***Significance at 1%.

fertilizers were more popular among male headed households compared to female headed households (Table 23). However, Table 23 results does not account for intensive margins. To address this limitation, Tables 24, 25 and 26 show results on area under SI technologies by research group, district, and gender and age category, respectively.

Table 24 results show that more land was allocated to crops under CA, inorganic fertilizer and organic manure. Unlike CA, farmers from the treatment group had more land under intercropping than the control group. As mentioned earlier, 49% of the sampled farmers

Table 24: Area under sustainable intensification technologies by research group

	Full sample (N=1,269) (1)	Treatment (N=948) (2)	Control (N=321) (3)	Mean difference (4) = (2) – (3)
Conservation agriculture (ha)	0.86	0.82	0.99	-0.16*
	(1.48)	(1.56)	(1.21)	
Inorganic fertilizer (ha)	0.81	0.77	0.92	-0.15
	(1.45)	(1.51)	(1.23)	
Organic manure (ha)	0.46	0.47	0.45	0.02
	(1.32)	(1.44)	(0.83)	
Intercropping (ha)	0.26	0.28	0.20	0.08**
	(0.48)	(0.51)	(0.39)	

Note: Mean values are shown with standard deviations in parentheses. N, Number of observations (No. of farming households). *Significance at 10%, **Significance at 5%.

Table 25: Area under sustainable intensification technologies by district

	Balaka (N=320) (1)	Dedza (N=194) (2)	Kasungu (N=232) (3)	Mangochi (N=222) (4)	Mzimba (N=172) (5)	Zomba (N=129) (6)
Conservation agriculture (ha)	0.91	0.57	1.49	0.40	1.02	0.67
	(2.35)	(0.63)	(1.48)	(0.43)	(1.02)	(0.64)
Inorganic fertilizer (ha)	0.70	0.46	1.43	0.48	1.11	0.66
	(2.26)	(0.67)	(1.52)	(0.40)	(0.97)	(0.67)
Organic manure (ha)	0.69	0.38	0.46	0.40	0.22	0.44
	(2.32)	(0.58)	(0.96)	(0.44)	(0.65)	(0.61)
Intercropping (ha)	0.31	0.33	0.11	0.29	0.08	0.47
	(0.69)	(0.36)	(0.38)	(0.31)	(0.27)	(0.51)

Note: Mean values are shown with standard deviations in parentheses.

Table 26: Area under sustainable intensification technologies by gender and age category

	Gender		Age category	
	Male HH (N=850) (1)	Female HH (N=419) (2)	Youth HH (N=364) (3)	Non-Youth HH (N=905) (4)
Conservation agriculture (ha)	0.94***	0.71	0.71**	0.93
	(1.06)	(2.08)	(2.10)	(1.13)
Inorganic fertilizer (ha	0.90***	0.63	0.65**	0.87
	(1.08)	(1.98)	(2.11)	(1.07)
Organic manure (ha)	0.46	0.48	0.40	0.49
	(0.73)	(2.04)	(2.08)	(0.83)
Intercropping (ha)	0.25	0.27	0.20***	0.28
	(0.40)	(0.62)	(0.34)	(0.53)

Note: Mean values are shown with standard deviations in parentheses. We used 35 years as cut-off point to define youth (<36 years) and non-youth (>35 years) categories. We used *t*-test to test the mean difference between Male HH and Female HH, and Youth HH and Non-Youth HH. **Significance at 5%, ***Significance at 1%.

had adopted intercropping. This is so because most farmers have limited arable land, as such intercropping or mixed farming is a coping mechanism to land scarcity.

Does land availability in the study districts play a significant role on explaining area under SI technologies? Table 25 show area under SI technologies in the six study districts. As expected, farmers in Kasungu and Mzimba allocated more land to SI technologies such as CA (i.e., crop rotation and zero/minimum tillage) and inorganic fertilizer. Probably, the population in the two districts is low relative to availability of arable land. Similar studies (e.g., Ricker-Gilbert et al., 2014; United Nation, 2015; Willy et al., 2019), also found that areas with lower population density are associated with bigger farm sizes.

Related to this point, farmers in districts with higher population density such as Mangochi and Balaka are associated with smaller farm sizes. As a result, more land is under intercropping in these districts, as a coping mechanism to land scarcity. Moreover, female headed households and the youth had allocated less land to SI technologies than male headed households and non-youth (Table 26). These results suggest that women, the youth and other marginalized groups continue to struggle to access productive assets including land. This could be key in improving resilience to climate change for millions of smallholder farmers, reducing GHG emissions, and enhancing environmental health and biodiversity.

5 | Conclusion

To summarize the key findings, most of the sampled households are Christian, male-headed, and involved in crop production. Most farmers allocate their land to maize and legume crops such as groundnut, soybean, and pigeon peas. However, almost half (49%) of sampled farming households adopted intercropping and only 35% used mixed farming system technologies.

Despite most farmers using inorganic fertilizer (75%), crop rotation (53%) and organic manure (49%) as well as improved seed varieties (44%), crop yields were marginally low, with maximum value of 2,073kg/ha. This echoes most prior studies (e.g., Benson, 2021; FAO, 2021; Khonje et al., 2022b) who found that crop yields, including cereals and legumes, have remained low (<3.5 tons/ha) in Malawi. On the contrary, the use of inoculants and agricultural lime is not popular among smallholder farmers. To keep their farm produce, most farming households use woollen and hermetic bags.

In addition to crop production, livestock farming is also a prominent farming activity especially for animals such as chickens and goats. This is a welcome development, since promoting livestock under the SI-MFS initiative could play a significant complementary role with crop-based farming systems and household resilience. However, the benefits associated with mixed farming may not be maximized as this (mixed farming) practice is not common in the study districts. One major implication of this finding is that extra efforts should be made to promote crop-livestock systems that suit small landholdings. Furthermore, crop and livestock-based interventions could be promoted as a package and not in insolation, which is common for most development projects in the Global South.

Regarding dwelling characteristics, majority of the households used burned bricks as the primary materials for walls and mud bricks to make floor. Most houses were roofed with leaves or grass. This is consistent with the aggregate wealth index data for the sampled households, where majority of the them were classified as poor. Probably, this is worsened by major recurring agricultural shocks including food inflation (68%), rising agricultural input prices (67%), drought (53%) and crop pest and disease outbreaks (42%) in the six study districts. Moreover, about 85% of the sampled households accessed drinking water from boreholes and not piped water. On other water, sanitation and hygiene characteristics, majority (54%) of the households used their own (private) traditional latrines with a wall and roof.

Several key issues are emerging from our analysis on the five impact areas: (1) nutrition, health and food security; (2) poverty reduction, livelihoods and jobs; (3) gender equality, youth and social inclusion; (4) climate adaptation and mitigation; and (5) environmental health and biodiversity. First, we generally observe that a farming household had FCS of 41 and HDDS of seven. To be precise, farming households from the treatment group had higher food consumption expenditure than those in the control group by MWK18,158. We also found that female headed households have lower scores for FCS (41.82 vs. 38.8) and HDDS (7.36 vs. 6.91) than male headed households. Young people have a higher HDDS and food consumption expenditure than adults.

Second, even though most of the sampled households have a diversified diet (they consumed seven food groups on average), poverty levels are still high (73%). This suggests that most farming households may still struggle to access highly nutritious food from food markets as well as accessing basic needs to support their livelihood. More poor households came from the control group (76%) compared to the treatment group (72%). Dedza had the highest household income than any other study district. However, women and the youths are earning less crop, livestock and non-farm income from farming and non-farm activities. Perhaps, these results entail that women farmers and the youths could be deliberately targeted as beneficiaries for the SI-MFS initiative.

Third, in addition to gender disparity on income and assets ownership including land, gender pay gap also exists on agricultural activities. Overall, male casual laborers earned more income than female casual workers when they were engaged in farming activities, non-farm activities and businesses in the study districts. Similarly, we observed a considerable gender disparity on time use and participation in community organizations. Male farmers mainly participated in community organizations such as church or mosque (47%), local administration (11%) and groups associated with input supply (11%). Whereas female farmers generally participated in village savings and loan associations (VSLA) (45%) and farmers' associations (9%). In general, these results underscore that gender and social inequalities are a persistent feature of agri-food systems including MFS and if not addressed, achieving the SDGs by 2030 will be difficult.

Lastly, with recurrent extreme weather events, achieving the SDGs without climate-sensitive interventions will be a formidable challenge. The results show that more land was allocated to crops under CA, inorganic fertilizer and organic manure. Our analysis suggest that population plays a contrasting role on allocation of land to SI technologies. Farmers in Kasungu and Mzimba—abundant land with less population—districts allocated more land to SI technologies such as CA (i.e., crop rotation and zero/minimum tillage). In contrast, farmers in districts with higher population density such as Mangochi and Balaka are associated with smaller farm sizes. As a result, more land is under intercropping in these districts, which may act as a coping mechanism to land scarcity.

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